MILITARY SPECIFICATION

TRANSMITTER, RADIO, AN/FRT-83(B)
1000 WATTS PEAK ENVELOPE POWER AND AVERAGE, HIGH-FREQUENCY, INDEPENDENT SIDEBAND

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction, and performance of the Radio Transmitter, AN/FRT-83(B), hereinafter referred to as the transmitter. The transmitter is a Navy general-purpose transmitter accepted for use on high-frequency (HF), independent sideband (ISB), and single sideband (SSB) radio communication circuits. The transmitter is a 1000 watt (w) peak-envelope-power (PEP) and average (avg) power, frequency synthesized, forced air-cooled radio transmitter that features complete automatic tuning capability to the frequency selected at the MD-777(A)/FRT, hereinafter referred to as the exciter. The transmitter is capable of remote control when equipped with the AN/FSQ-98(A) remote control system. The transmitter features modular construction that permits removal and replacement of subassemblies by maintenance personnel. The transmitter operating frequency range is from 2.0000 megahertz (MHz) to 29.9999 MHz in 100 hertz (Hz) steps.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-E-15090	Enamel Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of
MIL-S-19500/420	Semiconductor Device, Diode, Silicon, Power Rectifier, General Purpose Tx And Non-Tx Types 1N5550 Through 1N5554
MIL-E-21981	Electronics Type Designations, Identification Plates And Markings: Requirements For
MIL-C-28806	Coupler Group, Antenna, AN/URA-38( )
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification For
MIL-R-83725/4	Relays, Vacuum, SPDT, 3 Amperes, DC Load Switching 2.5 Kilovolts DC

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-449	Radio Frequency Spectrum Characteristics, Measurement Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

2.1.2 Other Government drawing and publication. The following other Government drawing and publication form a part of this specification to the extent specified herein.

DRAWING

NAVAL ELECTRONIC SYSTEMS COMMAND (NAVELEX)

01003150	Standard Plan Radio Transmitter AN/FRT-83(V)

PUBLICATION

NAVELEX

N00039-79-C-0165	Radio Transmitter AN/FRT-84( )

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

S1.4-1971	American National Standards For Sound Level Meters

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

152-1953	Recommended Practices For Volume Measurements Of Electrical Speech And Program Waves

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70-1983	National Electrical Code

(Application for copies should be addressed to National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels And Associated Equipment

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The transmitter shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Transmitter description. The transmitter shall be frequency-synthesized, ISB, forced air-cooled, and shall be designed for fixed station or transportable use. The transmitter shall consist of an Interconnection Box, J2821( )/FRT-83(B); an Amplifier, Radio Frequency, AM-6046( )/FRT-83(B); a Power Supply, PP-6067( )/FRT; a Modulator-Synthesizer, MD-777(A)/FRT; and a Frequency Shift Keyer, KY-655(A)/FRT. All of these components shall be contained within one steel equipment cabinet. The exciter shall determine the frequency of the output signal, provide the modulation process, determine the output frequency stability, and furnish an output signal of at least 250 milliwatts (mW) at 50-ohms nominal impedance to the radio frequency (RF) amplifier of the transmitter. Blank panel space shall be provided in the transmitter as shown in NAVELEX Drawing (DWG) 0100315D.

3.2.1 Transmitter dimensions. The complete transmitter in one cabinet, less shock mounts, shall have overall dimensions specified in a through c:

a. Height: 199 centimeters (cm) (78.313 inches (in.))
b. Width: 54 cm (21.25 in.)
c. Depth: 107.3 cm (42.25 in.)

The cabinet shall contain the J2821( )/FRT-83(B), the AM-6046( )/FRT, the PP-6067( )/FRT, the MD-777(A)/FRT, and the KY-655(A)/FRT and, when the transmitter is so equipped, the local control interface unit (LCIU) of the AN/FSQ-98(V) remote control system. The transmitter configuration shall be in accordance with NAVELEX DWG 0100315D.

3.2.2 Retractable cable harness. Cable harnesses associated with the control-indicator drawer assembly and the modulator-synthesizer MD-777(A)/FRT shall be equipped with a flat-pack-type (molded) cable harness and retractable carrier assembly. The carrier assembly shall be constructed to preclude damage to the cable harness during all conditions of assembly, removal, insertion, or tilting (±90 degrees).

3.2.2.1 Cables. Cables supplied with the end items shall be fabricated in accordance with Requirement 66 of MIL-STD-454 and tested as specified in 4.5.3.

3.2.3 Accessibility. The transmitter shall be so constructed that parts, terminals, and wiring are accessible for circuit-checking and adjustment, maintenance, repair, and replacement with a minimum number of commercial or special tools (see 6.7). Parts shall be mounted so that their identification marking will be readily visible with minimum disassembly of the transmitter. All required special tools shall be provided with each transmitter.

3.2.4 Ventilation and cooling. The transmitter shall be ventilated and cooled in such a manner that under normal conditions of operation at the maximum ambient temperature of 50° Celsius (C), no part shall reach a temperature which will cause damage. Forced air-cooling with replaceable air filters provided at each air inlet shall be used. Filters shall be removable and replaceable while the transmitter is in operation without affecting normal operation or endangering operating personnel. A cooling air inlet shall be located on the front of the amplifier.

3.2.4.1 Ducting. The power amplifier (PA) shall be fitted for ducting of the exhaust air.

3.2.5 Acoustic noise. The transmitter shall produce no more than 65 decibels (dB) of acoustic noise measured by instrumentation having an A weighting characteristic in accordance with MIL-STD-1472 (A weighting, ANSI S1.4-1971) at a distance of 3 meters (m) (10 feet (ft)) in any direction normal to the nearest outer surface of the transmitter when it is installed and operated in a large transmitter room without special acoustic treatment and with an ambient noise level not greater than 50 dB. (Reference noise level is a free progressive 1000-cycle sound wave of 10^{-16} watts per square centimeter (W/cm^2)). Acoustic treatment of the AM-6046( )/FRT-83(B), cabinet walls, and the door is permitted if necessary to conform to this requirement.

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3.2.6 Insulating compounds. Compounds which soften and flow at any temperature less than +75°C shall not be used unless contained in such a manner that the compound will not flow from the container during service use. Compounds which crack at any temperature above -62°C shall not be used.

3.2.7 Safety to personnel. Provision shall be made to prevent personnel from contacting any potential in excess of 50 volts root-mean-square (Vrms). All compartments in which this potential is exceeded shall be enclosed and protected with interlocks on all access doors or panels which may be opened during operation or maintenance. Interlocks shall act to remove all potentials within the protected compartment which are in excess of 50 Vrms. In cases where the removal of a potential is impractical, such as line input terminals, the terminals shall be insulated and protected with a nonmetallic cover labeled CAUTION and stating the voltage. The transmitter interlock circuit shall have a provision for an external extension. To this end, a pair of terminals shall be provided from which the interlock circuit may be extended for at least 61 m (200 ft), using the same sized conductors as used within the internal interlock circuit. Interlock voltage shall not exceed 50 Vrms. Transmitter design shall conform to the Safety criteria paragraph of MIL-E-16400 except that leakage current shall not exceed 5 milliamperes (mA) and shall be measured in accordance with 4.5.4. Interlock indicators shall be provided as required by the Interlock indicator paragraph of MIL-E-16400.

3.2.7.1 Capacitor discharge. Provisions shall be made to prevent access to any compartment of the transmitter without activation of a mechanical device to discharge any capacitor in which the product of the capacitance, in farads, and the square of the operating voltage, in volts (V), exceeds 25, except that such device need not be provided for a compartment where the maximum voltage in the compartment does not exceed 50 Vrms.

3.2.7.2 Shorting stick. An electrical shorting stick with readily accessible stowage and cable of sufficient length to reach all high voltage capacitors used anywhere in the transmitter shall be provided. Cable shall be permanently connected to a ground point having less than 0.01 ohm resistance. Breakdown voltage for the handle shall be at least 10 times the maximum peak voltage encountered.

3.2.7.3 Weight marking. Transmitters weighing in excess of 18.2 kilograms (kg) (40 pounds (lbs)) shall have the weight clearly identified on the external surface of the transmitter and readily visible during installation and removal.

3.2.7.4 Electrical power ON-OFF switches. A switch for disconnecting the transmitter from each of the electrical power systems, including the 115 volts alternating current (VAC) or 230 VAC primary source of 3.4.11.1, shall be mounted on the circuit breaker panel and its function clearly labeled. The switch shall break all power conductors of the power circuit.

3.2.7.4.1 Power indicator lamp. A green power indicator lamp shall be mounted near each of the power switches to indicate when the equipment is energized. The lamp shall be connected to the load side of the switch and across the input power conductors.

3.2.7.5 Overcurrent device location. A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded conductor as determined by the National Electrical Code (NFPA 70-1983) for grounded source power. The overcurrent device shall be connected to the load side of the power switch. If circuit breakers are used, they shall also serve as the main power switch. Circuit breakers shall provide a visible indication of status when tripped.

3.2.8 Protection of internal components. All major components of the transmitter shall, upon failure, activate protective features preventing further damage. All electrical indicating instruments with associated multipliers or transformers, except self-contained thermocouple instruments, shall be protected by the use of by-pass units against damage due to stray RF currents or transient potentials. All delicate instruments and devices shall be mounted for protection from damage due to shock and vibration.

3.2.8.1 Transmitter protection. The design and construction shall be of such character that damage to the transmitter shall not occur from operator error such as improper control sequencing, improper tuning, failure to respond to alarms or fault indicators, and failure to follow operating instructions.

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3.2.8.2 External circuit fault protection. External circuit fault protection which conforms to the requirements of the Electrical tests paragraph of MIL-E-16400 shall be provided. All components shall be protected from damage in the event of failure or malfunction of any external circuit, including prime power, audio input sources, output coaxial line, or other portions of the load circuit. Failures or malfunctions of external circuits shall include all possible types, such as open circuit, short circuit, intermittent open and short circuits, excessive voltage standing wave ratio (VSWR), long term overvoltage and undervoltage of prime power, improper phase rotation, loss of one or more phases, and transients. For definition of long term overvoltage and undervoltage see 6.4.1.

3.2.8.3 VSWR protection. The design of the transmitter shall be such that operation into a load which generates a 4.0:1 VSWR shall not cause damage.

3.2.8.4 Internal transient voltage protection. Internal transient voltage protection circuitry shall be incorporated into the design of the transmitter. The circuitry shall include, but shall not necessarily be limited to, protection from transient voltages generated by transmitter shut-downs, circuit breaker activation, circuit breaker deactivation, change of frequency, changing modes of operation, proximity of cabling, sudden short or open in transmission output line, loss of main alternating current (AC) input power on any combination of phase or phases, any form of arcing, and any other transient voltage condition. Provision of transient voltage protective circuitry or other corrective measures, such as selection of a different component or element, shall be accomplished to reduce conditions to a point where the voltage rating of any component or element is not exceeded by a voltage more than two times the voltage rating for a period greater than 2 milliseconds (ms).

3.2.8.5 Power supply protection. Fault conditions ranging from open circuits to short circuits shall cause no damage to the power supply.

3.2.8.6 Printed circuit (PC) assembly protection. With power applied to the transmitter, removal or insertion of PC assemblies shall not damage or degrade any part of the transmitter. Each assembly shall be keyed to prevent insertion into the wrong connector and to prevent reversal of insertion.

3.2.9 Test points. The transmitter shall be provided with adequate and well-defined test points to facilitate making performance measurements and for use in maintaining the equipment. Test points and test features shall be in accordance with the applicable requirements of MIL-STD-415.

3.2.9.1 Test jacks. Test jacks provided in accordance with 3.2.9 and subject to the influence of interference shall be equipped so that shielded test leads can be used. When a test jack is to be used with a spectrum analyzer, the output of the test jack shall be within the range of -50 decibels referred to one milliwatt (dBm) to 0 dBm. All RF sampling voltages shall be 1 Vrms based on a 1 kilowatt (kW) PEP output when modulated with a two-toned signal. A jack shall be provided to insert the output of a two-tone oscillator at each audio input of the transmitter.

3.2.9.2 Test cords and module extenders. All test cords, PC cards, or module extenders required for operation and maintenance of the transmitter shall be provided with the equipment. All modules which make up the major components of the exciter and amplifier shall be completely removable by a single disconnect. Signal and power for servicing and testing of modules in the extended position shall be provided by the basic unit.

3.2.9.3 General purpose test equipment (GPTE). GPTE referenced in all technical documentation (manuals, training aids, and so forth) shall be the same as specified in the Government-furnished equipment (GFE) manuals.

3.2.10 Meters and indicators. Meters shall be provided to indicate all critical circuit parameters where the function indicated is of the continuously variable type, such as plate current, output power, bias voltage, and exciter output. Where not otherwise specified, meters shall be of an accuracy of 5 percent or better.

3.2.10.1 Multi-test meters. Meters shall be provided in the exciter and the 1-kW amplifier and shall be capable of being switched to any circuit which does not require continuous monitoring, but does require periodic adjustment or monitoring inspection.

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3.2.10.2 Elapsed-time meters. An elapsed-time meter shall be provided in the transmitter to indicate operating hours, defined as high voltage ON time. A meter to record filament hours for the final PA electron tubes shall be provided.

3.2.10.3 Tune cycle counter. A digital recording counter shall be provided to record the number of tune cycles. This feature shall apply to the automatic and manual tuning modes. This counter shall be capable of recording and displaying in an unambiguous manner up to 9999 events and shall be capable of being reset to 0. The reset feature shall be suitably protected so that the counter cannot be inadvertently reset. The counter shall be mounted so that access may be easily obtained without interrupting service.

3.2.11 Controls. All transmitter operational controls shall be located on the front panels and be so grouped, arranged, and integrated by function with the applicable meters and indicators that an orderly, simple, and efficient method of performing normal operating functions may be used by a single operator.

3.2.11.1 Marking. Words or phrases descriptive of the control functions shall be marked adjacent to each operating control. Markings shall be permanent and of such size as to be readable by a person with normal vision at a distance of 60 cm (24 in.) with an illumination as low as 10.8 lux (one foot-candle).

3.2.11.2 Manual override. Override of the automatic tuning shall be provided so the transmitter may be manually tuned.

3.2.12 Interchangeability. Like units, modules, PC cards, assemblies, subassemblies, and replaceable parts shall be physically, electrically, and functionally interchangeable with corresponding units produced in accordance with the requirements of this specification and the AN/FRT-83A or the AN/FRT-83(V) provided as GFE (see 6.6). Individual items shall not be hand-picked for fit or performance. Interchangeability requirements shall be two-way, except as specified in 3.2.12.1, 3.2.12.2, and 3.2.12.6.

3.2.12.1 Exciter interchangeability. The exciter, produced in accordance with the requirements of this specification, shall be completely interchangeable with like items in the AN/FRT-83(V), the AN/FRT-83A, the AN/FRT-84(V), the AN/FRT-84A, the AN/FRT-84( ) procured under contract N00039-79-C-0165, and the AN/FRT-85(V). The AN/FRT-83A or the AN/FRT-83(V) provided as GFE shall be interchangeable with corresponding items produced in accordance with the requirements of this specification, except for two ISB operations in accordance with 3.4.12.5.4.

3.2.12.2 AM-6064( )/FRT and PP-6067( )/FRT interchangeability. The AM-6064/FRT and the PP-6067( )/FRT produced to conform to the requirements specified herein shall be interchangeable with like components in the AN/FRT-83(V), the AN/FRT-83A, the AN/FRT-84(V), the AN/FRT-84A, and the AN/FRT-84( ), except as specified in 3.4.14.2 and 3.4.14.2.5.

3.2.12.3 KY-655( )/FRT interchangeability. The KY-655( )/FRT keyer produced to conform to the requirements specified herein shall be interchangeable with the KY-655/FRT provided as a component of each of the transmitters listed in 3.2.12.1.

3.2.12.4 AN/FSQ-98(V) LCIU interchangeability. The transmitter shall be designed to permit installation of the LCIU of the AN/FSQ-98(V) and operation under the remote control of the remote control unit (RCU) of the AN/FSQ-98(V). No engineering changes or modifications to the transmitter shall be necessary to conform to this requirement. LCIU installation shall be accomplished by making simple cable connections without soldering or the use of cable connector adaptors. The transmitter shall be equipped with a blank panel which can be mounted in the transmitter equipment rack to cover the void which exists when the LCIU is not installed.

3.2.12.5 RF tuner interchangeability. The transmitter shall be designed to be backfitted with the RF tuner, the TN-509/FRT-83(V), the MFR1 14304, the PN6049-4500 and the Tuner Control, C-7704/FRT-83(V), MFR 14304, PN6049-4000, both of which are included in the AN/FRT-83(V), and the AN/FRT-83A. Backfitting shall be possible with no engineering changes or modifications to the transmitter and shall be accomplished, if required, by mounting the tuner and the tuner control in the transmitter equipment rack and by making simple cable connections without soldering or the use of cable connector adaptors. The transmitter equipment rack shall be predrilled as required to mount the tuner and the tuner control in the same manner as the GFE.

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3.2.12.5.1 Tuner emulator panel. The transmitter shall be equipped with a tuner emulator panel which can be mounted to cover the voids resulting from not installing the tuner and its control. The tuner emulator panel shall include components which perform all the tune logic functions provided by the RF tuner, when installed, and the final RF power output metering of 3.4.14.10. Insertion loss for the tuner emulator panel shall not exceed 0.5 dB. Electrical and mechanical interfaces for the tuner emulator panel shall be identical to the interfaces provided for the RF tuner installed in prior transmitters.

3.2.12.6 AN/FRT-84 driver amplifier interchangeability. Components of the AN/FRT-83(V) and the AN/FRT-83A transmitters are used to drive the AM-6047/FRT final PA of the AN/FRT-84(V), the AN/FRT-84A, and the AN/FRT-84(B) specified in 3.2.12.1. Corresponding components of the transmitter produced to conform to the requirements specified herein shall be interchangeable with those components of the AN/FRT-83(V) and the AN/FRT-83A and shall function as a driver for the AM-6047/FRT without engineering changes or modifications, except as specified in 3.2.12.1 and 3.2.12.2.

3.2.13 Serial numbers and nomenclature. All major components of the transmitter shall have serial numbers and nomenclature. Nomenclature shall be requested in accordance with MIL-E-21981.

3.3 Construction. Construction shall be as specified in 3.3.1 through 3.3.6.

3.3.1 Cabinet construction. The transmitter cabinet and its internal panels and plates shall be constructed of material of sufficient strength to provide a rugged and durable enclosure adequate for either fixed-station or transportable service. Construction of the cabinet shall be consistent with the requirements of 3.5.3.

3.3.1.1 Side panel removal. The cabinet shall be assembled with captive nuts or nut bars to permit removal of damaged side panels without the necessity for total disassembly (or replacement) of the individual cabinet.

3.3.1.2 Cabinet door and fasteners. Components in the transmitter cabinet shall be accessible from the rear via a hinged door. The door hinge(s) shall be either full-length and continuous or, at a minimum, positioned at three points. Door fasteners (latches) shall be of the quarter-turn handle design.

3.3.2 Degree of enclosure. The degree of enclosure shall be the same as the GFE transmitter. Front access shall be provided for normal repair and maintenance. Cabinet rigidity shall be sufficient to survive the shock test specified in 3.5.3 without deforming. Openings which would otherwise admit rodents, insects, or dirt shall be screened or equipped with replaceable air filters.

3.3.3 Mountings. All major units shall be mounted on slides, tracks, or hinges to permit servicing of individual units from the front of the transmitter. For all units weighing less than 22.7 kg (50 lbs), the mounting shall also provide for a +90 degree and a -90 degree front-to-back tilting to a horizontal position in increments of 45 degrees with provisions for locking (hold) at each 45-degree step. The -90 degree tilting requirement shall not apply to the bottom slide or track-mounted units. Major units not mounted on slides or tracks shall be readily accessible from the front of the transmitter for servicing purposes. The KY-655/FRT shall be tiltable only to the +45 degree and -45 degree positions.

3.3.3.1 Support strength. Drawer slides, detents, mounting surfaces, and other attachment mechanisms which support equipment shall be tested to twice the weight for the equipment supported.

3.3.3.2 Tip stability. Design of rack-mounted equipment shall maintain the center of gravity as low as possible.

3.3.3.3 Drawer slide security. Equipment design shall include provisions to prevent accidental derailing and pulling out of equipment mounted on drawer slides.

3.3.3.4 Equipment latch security. Exterior doors, hinged covers, and drawers shall have automatic latch or quick release devices which must be actuated before they can be opened and stops to hold them in the open position for each maintenance configuration.

3.3.4 Modular construction. The use of modular construction shall be the same as in the GFE.

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3.3.5 Finish. Exterior cabinet surfaces shall be finished with enamel conforming to Type III, Class 2 of MIL-E-15090.

3.3.6 Unacceptable materials. The use of unacceptable materials shall be restricted in accordance with the Unacceptable material paragraph of MIL-E-16400.

3.4 Performance characteristics. The performance characteristics shall be as specified in 3.4.1 through 3.4.15.4.2.2.

3.4.1 Frequency range. The frequency range of the transmitter shall be incrementally continuous from 2.0000 MHz through 29.9999 MHz in increments of 100 Hz.

3.4.2 Frequency stability. The frequency stability of the entire transmitter shall be determined by the exciter's 1-MHz internal standard having a stability which shall be no worse than 1 part in 10^8 when measured at the RF power output point during any 24-hour period at any frequency covered by this transmitter. The absolute frequency error shall have a daily change of not more than 0.3 Hz. The root-mean-square (rms) phase deviation shall not exceed 3 degrees in 10 ms.

3.4.3 Frequency selection. The frequency selection for the transmitter shall be entirely a function of the exciter. The absolute frequency of the RF carrier, whether present in the transmitter output or fully suppressed, shall be determined by the exciter and shall be selected by manual controls on the front panel of the exciter or by the remote control system, AN/FSQ-98, when so equipped.

3.4.4 Operating modes. The transmitter shall provide the modes of operation or types of emission specified in a through k:

a. AØ: Absence of any modulation
b. A1: Continuous wave (CW) telegraphy
c. A2: Modulated CW telegraphy
d. A3E: Compatible amplitude modulated telephony
e. A3A: SSB, reduced carrier (upper sideband (USB) or lower sideband (LSB))
f. A3B: Two ISBs
g. A3J: SSB, suppressed carrier
h. A9B: Two ISBs with combined telephony and telegraphy
i. A9J: Four ISBs with combined telephony and telegraphy
j. F1: Frequency shift keying (FSK)
k. F4: Facsimile (FAX)

3.4.5 Electrical noise. For definition of electrical noise see 6.4.2. Tolerance: the electrical noise of the transmitter shall be at least 60 dB below the transmitter's full rated 1-kW avg power output reference.

3.4.5.1 Key-up. The transmitter shall be capable of being employed in simplex operation without causing electrical interference to the corresponding radio receiver frequency while the transmitter is in key-up or receive-mode condition. When the transmitter is in the key-up or receive-mode condition, the transmitter avg noise power output measured in a 3000-Hz bandwidth anywhere between 2.0000 MHz and 29.9999 MHz shall be at least 115 dB below 1 mW.

3.4.5.2 Intelligence loss. When the transmitter is changed from key-up or receive-mode condition to key-down condition, not more than 8 ms of intelligence shall be lost.

3.4.5.3 Dry contact keying. A pair of terminals and circuitry shall be provided in the exciter rack for dry contact ON-OFF (or push-to-talk (PTT)) keying in the USB mode of the exciter and the amplifier's key-up noise reduction circuitry during simplex operation. A keying terminal shall be provided with a means for a circuit closure when external PTT control of the transmitter is not required.

3.4.6 Out-of-band spurious emission. The power level of all out-of-band spurious emissions below the level of the unsuppressed level of the carrier frequency, including harmonic distortion, parasitic oscillations and intermodulation products outside of the normally transmitted bandwidth of 12.6 kilohertz (kHz), shall be as specified in a and b:

a. Any emission appearing on any frequency removed from the unsuppressed level of the carrier frequency by at least 50 percent but not more than 100 percent of the maximum authorized bandwidth shall be attenuated not less than 43 dB.

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b. Any spurious emission appearing on any frequency removed from the unsuppressed level of the carrier frequency by at least 100 percent but not more than 300 percent of the maximum authorized bandwidth shall be attenuated not less than 80 dB.

3.4.7 In-band intermodulation distortion. All in-band intermodulation products shall be measured from the audio inputs to the RF output of the transmitter. Requirements specified herein apply to any operating frequency from 2.0000 MHz to 29.9999 MHz.

3.4.7.1 In-band intermodulation distortion when white noise loaded. When channels B_2, B_1, and A_2 are loaded with white noise signals which produce 200 W avg power into a 50-ohm load connected to the RF output point of the transmitter, the avg power level measured in the nominal 3-kHz spectrum of channel A_1 shall be at least 43 dB below the 200 W avg power output of the transmitter. When the avg power output level of the transmitter is to be measured using a frequency domain indicating device in conjunction with a calorimetric measuring set, a 4.77 dB factor derived from the formula:

$$dB = 10 \log \frac{P_1 + P_2 + P_4}{P_1}$$

may be added to the avg power difference between channel B_1 or any other loaded channel, and the avg power level P_3 appearing in channel A_1 to account for power integration in the 50-ohm termination resulting from equal power distribution in the three loaded channels. P_1, P_2, and P_4 are defined as:

P_1 = the avg power level appearing in the nominal 3-kHz spectrum of channel B_2 at the transmitter RF output point. The relationship between the total white noise loaded power output of the transmitter (200 W avg) and the avg power appearing in the RF spectrum of channel B_2 or any other loaded channel (B_1 or A_2) is:

$$P_1 = \frac{200}{3} = 66.66 \text{ W avg}$$

$$P_2 = P_1 \text{ and } P_4 = P_2$$

P_3 is the avg power appearing in the nominal 3-kHz spectrum of channel A_1 (the unloaded channel) at the transmitter RF output point when channels B_2, B_1, and A_2 are loaded with equal white noise signals such that 200 W avg power output will result (see FIGURE 1).

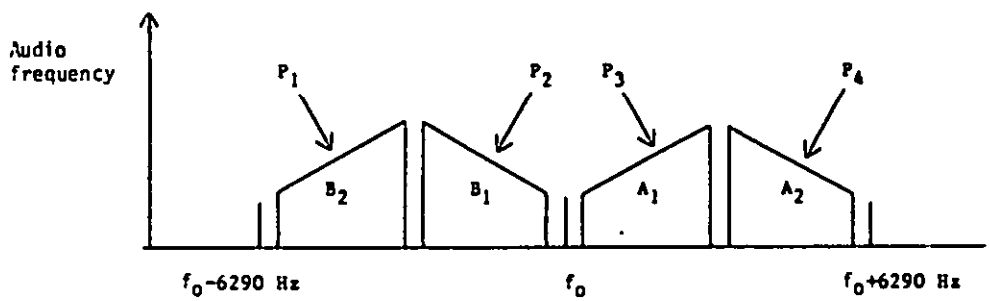


FIGURE 1. Transmitter audio frequency (AF) channels.

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3.4.7.2 Two-tone performance. When operated in the A3J mode at 1 kW PEP (500 W avg power) with two-tone signals at 1000 Hz and 1625 Hz applied to channel A (USB), or channel B_1 (LSB), in accordance with the requirements of 3.4.14.2.1 for full rated PEP (two-tone) output, all intermodulation distortion products shall be at least 35 dB below either tone appearing at the output point of the transmitter. This requirement shall also include band-edge performance.

3.4.7.3 In-band spurious emissions and hum products. No in-band spurious emission, including hum products, shall be less than 35 dB below the level produced by single tone modulation. When any nominal 3-kHz channel is modulated with white noise signals, the in-band spurious emission and hum products shall be not less than 43 dB below the avg power produced at the output of the transmitter.

3.4.8 Output impedance. The active output impedance of the transmitter shall be designed to operate into 50-ohms nominal. No degradation of specified performance shall occur when the external load impedance is such that it produces a VSWR of up to 2:1 at the transmitter output fitting. No damage shall occur to any component of the transmitter when the external VSWR approaches the limit of 4:1. The transmitter shall be fitted with a standard 50-ohm Type N connector in accordance with MIL-C-39012 at the RF output point.

3.4.9 Interference. The Procurement of equipments and subsystems having met other EMI requirements paragraph of MIL-STD-461 is applicable to this specification. The transmitter shall conform to the requirements of MIL-STD-461 for Class A3 equipment. Measurement of the RF signature shall be made in accordance with MIL-STD-449.

3.4.10 Electromagnetic field. The transmitter shall operate without derating or degradation of performance when immersed in an RF field of 10 kHz through 30 MHz, whose power equals 0.01 W/cm^2. See 3.2.8 for protection of tuning and conditioning circuits.

3.4.11 Power supply. The entire transmitter, with the exception of the exciter and keyer, shall be designed to take power from a single source at any of the line voltages specified herein ±10 percent. The choice of line voltage shall be available to the Navy installer. Internal transformer taps for each line voltage, ±10 percent, of 200 V, 208 V, 220 V, 230 V, 240 V, 440 V, 460 V, and 480 V, on each tap, 50 Hz or 60 Hz, ±5 percent, 3-phase, shall be provided to permit the choice of nominal line voltage. Input power to the transmitter shall be connected to a suitably rated circuit breaker upon entering the transmitter cabinet. The circuit breaker and any associated terminal strips on the line side of the circuit breaker shall be enclosed to prevent accidental contact by personnel when working in the cabinet. The exciter and keyer shall each accept power from a separate source at 115 V and 230 V, ±10 percent, 50 Hz or 60 Hz, ±5 percent, single-phase. The exciter and keyer input power circuits shall each be equipped with a toggle switch which permits selection of either 115 VAC or 230 VAC. A provision shall be made for an alternate transformer-coupling of power input to the exciter and keyer from the prime power source.

3.4.11.1 Exciter, keyer, and LCIU power distribution. Primary AC power for the exciter, keyer, and, when installed, the LCIU shall be derived from a separate 115 V or 230 V, single-phase, 50 Hz or 60 Hz, ±5 percent, AC power source. An emergency source of 115 VAC, independent of the 3-phase AC power input voltage, shall be derived from a single-phase of the polyphase transmitter AC power transformer. FIGURE 2 illustrates the concept for the provision of primary and emergency AC electrical power to the exciter keyer and LCIU. A set of taps on the primary source input shall provide a method of compensating for variations in input line voltage of ±20 V from the nominal value of 115 V or 230 V in steps of 5 V. Each tap shall be clearly labeled to indicate the value of input line voltage with which it is to be used. Output taps may be used, if necessary, to conform to the output voltages, 115 V and 230 V, specified. The primary power source shall provide an output of 115 V or 230 V, 50 Hz or 60 Hz, ±5 percent, single-phase, dependent upon the value of input line voltage, when the input line voltage to the primary circuit is within the ±20 V tolerance. Failure of the primary source of electrical power shall cause the switch to automatically select the emergency power source without interruption of service. A set of test points which are accessible when the components are withdrawn from the cabinet shall be provided to permit the use of GPTE to measure the voltage to the exciter, keyer, and LCIU. AC power to the exciter, the keyer, and the LCIU, when installed, shall be distributed by a standard multiple outlet strip using grounding type outlets.

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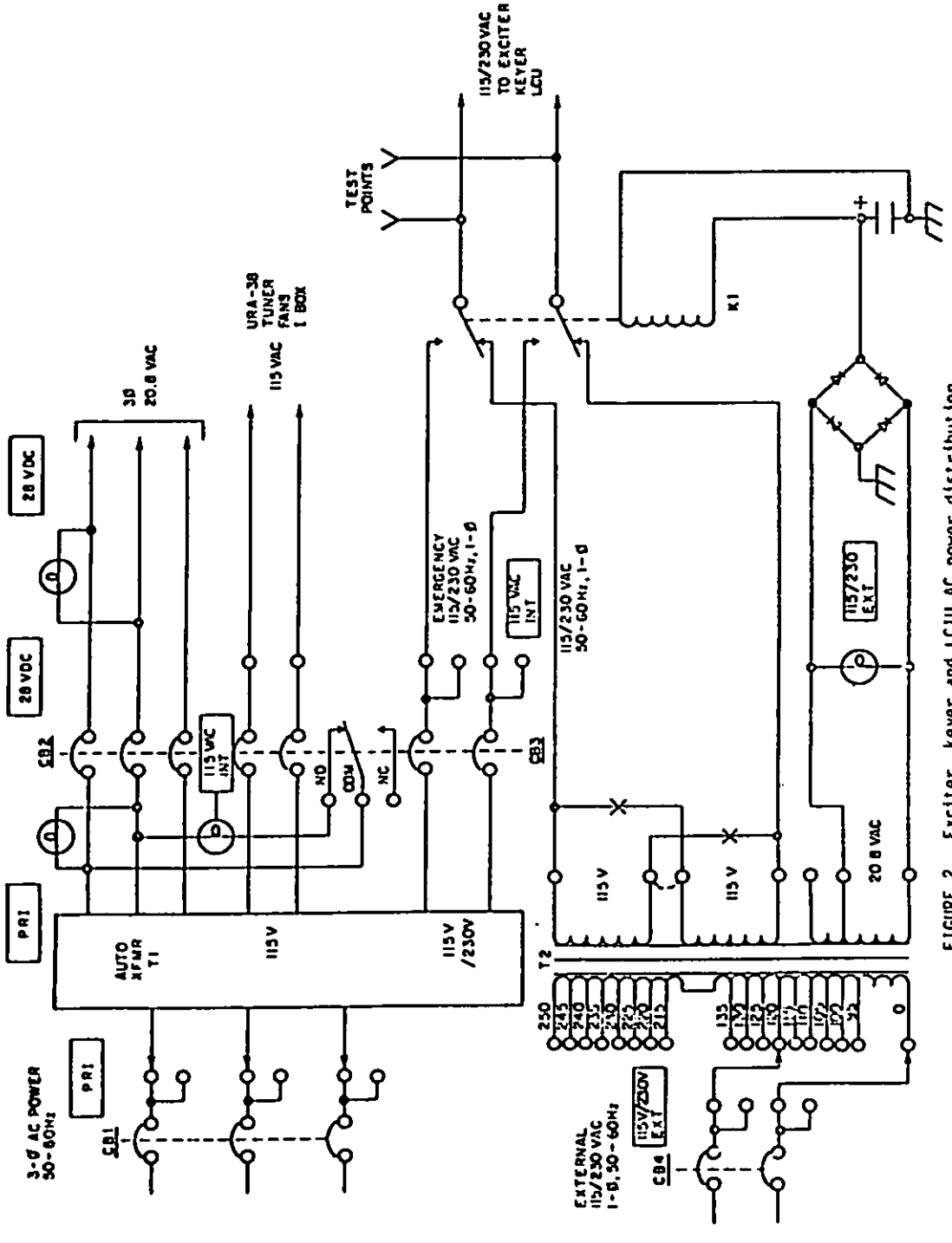


FIGURE 2. Exciter, keyer and LCIU AC power distribution.

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3.4.12 Exciter. The exciter requirements shall be as specified in 3.4.12.1 through 3.4.12.21.

3.4.12.1 Exciter characteristics. The exciter shall include all of the output frequency determining elements and all of the elements determining the output modulation characteristics. The exciter shall provide the termination for the audio input lines and shall furnish to the RF amplifier a modulated RF excitation signal adjustable between a minimum of not more than 10 mW PEP and a maximum of not less than 250 mW PEP throughout the frequency range of 2.0000 MHz to 29.9999 MHz in 100-Hz increments across 50-ohms nominal impedance. The exciter shall provide all required tuning and conditioning information to the amplifier. The exciter shall contain all of the input audio line terminations and input signal processing devices and circuitry. The exciter shall provide highly reliable and stable ISB operation consistent with the conditions specified herein. The exciter shall be frequency-synthesized, having self-contained provisions for accepting modulation from one, two, three, or four 3-kHz bandwidth audio input channels. The exciter output shall be capable of a maximum of not less than 250 mW PEP RF power in an A3J, A9B, or A9J operating mode depending upon the number of AF inputs utilized. VSWR shall not exceed 1.5:1 when terminated in 50-ohms resistive.

3.4.12.2 Frequency standards. The frequency standards shall be as specified in 3.4.12.2.1 through 3.4.12.2.7.1.

3.4.12.2.1 Internal frequency standard. The internal frequency standard shall have stability equal to that specified in 3.4.2. The internal frequency standard shall provide three separate 1-MHz outputs at a level of 1 Vrms when terminated into a 50-ohm load. Two of these three outputs shall be brought out to connectors on the front and rear panels of the exciter.

3.4.12.2.1.1 Rear panel connector. One coaxial connector of 50-ohms impedance marked 1 MHz INT STD MON shall be mounted on the rear panel of the exciter to allow connection via a trench duct. The connector shall be designed to mate with a standard 50-ohm Type BNC fitting in accordance with MIL-C-39012.

3.4.12.2.1.2 Front panel connector. One coaxial connector of 50-ohms impedance marked 1 MHz INT STD MON shall be mounted on the front panel of the exciter. The connector shall be designed to mate with a standard 50-ohm Type BNC fitting in accordance with MIL-C-39012.

3.4.12.2.2 Absolute frequency error. For definition of absolute frequency error see 6.4.3.

3.4.12.2.2.1 Aging. The absolute frequency error shall have a daily change of not more than 0.3 Hz, except during the initial aging period. (The initial aging period shall have been completed prior to the time that the transmitter is offered for Government tests.) The absolute frequency error shall not exceed the limits specified in a through c:

Time since last frequency calibration frequency calibration	Allowable absolute error at 29.9999 MHz at 29.9999 MHz
a. Less than 24 hours	0.3 Hz
b. Between 1 day and 7 days	0.3 Hz per day or 1.5 Hz, whichever is less
c. Between 7 days and 90 days	1.5 Hz per week or 3.0 Hz, whichever is less

3.4.12.2.3 External frequency standard. The exciter shall be provided with a means of connecting to an external 1-MHz prime frequency standard in lieu of the internal frequency standard. The exciter shall conform to performance requirements when using the external standard as long as the external standard delivers a signal equal to or better than the internal standard with respect to stability, amplitude, and purity. Connectors designed to mate with a standard 50-ohm Type BNC fitting shall be located on both the front and rear panels of the exciter to receive the 1-MHz external frequency standard signal. The level of this signal shall be 1 Vrms across a 50-ohm load. The connector located on the front panel shall be marked 1 MHz EXT STD MON; the connector located on the rear panel shall be marked 1 MHz STD IN.

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3.4.12.2.4 Phase-locking. The exciter shall be equipped with a manually adjustable memory type phase-locking circuit for locking the internal frequency standard to the 1-MHz external frequency standard. If either standard fails, the standard which is still in operation shall automatically become the exciter output frequency determining standard.

3.4.12.2.5 Automatic switching. A failure sensing modular device shall be provided to switch frequency control from the external to the internal standard upon failure of the external standard. Automatic switching shall occur when the amplitude of the external standard decreases 3 dB below a 1 Vrms reference level. After restoration of the external standard to at least 1 Vrms, the sensing device shall automatically switch back to the external frequency standard. The allowed transit time for the switching device to function shall not exceed 10 ms. No degradation to the transmitter output shall occur as a result of the 3 dB amplitude decrease at the output of the external standard.

3.4.12.2.6 Failure alarm. A front mounted visual failure alarm device with provisions for remoting to other locations shall be provided to indicate the loss of external frequency standard output. Terminals shall be provided in the transmitter for remoting this alarm feature.

3.4.12.2.7 Calibration controls. Controls shall be provided for adjustment of the frequency of the internal frequency standard of the exciter. Such controls shall have an adjustment range sufficient to allow reduction of the absolute frequency error as necessary once every 90 days for a period of at least 10 years. Controls shall be accessible for use while the exciter is in service but shall be so arranged or protected that inadvertent movement of any control is unlikely. Such controls shall have sufficient vernier spread to allow the minimum frequency increment to be small enough to reduce the absolute frequency error of the output carrier frequency to less than 0.3 Hz when the exciter is operating at any frequency between 2.0000 MHz and 29.9999 MHz. Such controls shall have a scale or markings which allows a log to be obtained for long term calibration record purposes. For definition of frequency calibration of the exciter see 6.4.4.

3.4.12.2.7.1 Monitoring output. The internal frequency standard signal present at the coaxial output specified in 3.4.12.2.3 may be used for calibration. Any absolute frequency error in the exciter output frequency shall have a simple arithmetic relationship to the absolute error of the 1-MHz output signal.

3.4.12.3 Frequency controls. The control and display of the selected frequency shall be functions of one device. The selected frequency display shall be located on the exciter front panel, shall be illuminated from the rear, and shall use Arabic numerals to indicate which of its possible digits has been selected. The controls shall be so constructed that there is no possibility of the exciter carrier frequency being other than that which is indicated. The size of the numerals and their illumination shall be such that they may be easily read by a person with normal vision at a distance of 61 cm (24 in.) when illuminated at a level of 10.8 lux (1 foot-candle).

3.4.12.3.1 Number and function of controls. There shall be one digital, decimal control for each digit of the frequency to be selected. This requires six controls, one for each of a through f:

a. Tens of MHz
b. Units of MHz
c. Hundreds of kHz
d. Tens of kHz
e. Units of kHz
f. Tenths of kHz

These controls shall be arranged in a horizontal line reading from left to right in conventional form.

3.4.12.4 Operating modes. The exciter shall provide modes of emission as specified in 3.4.4. A control marked SIDEBAND SELECTOR and a control marked CLASS OF EMISSION shall be provided on the exciter front panel. These controls shall be in the form of rotary selector switches.

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3.4.12.4.1 Sideband selector control. The sideband selector control shall provide selection of the sideband operating combinations specified in a through f:

a. USB (channel A_1) with voice-frequency-gate (VFG)
b. USB (channel B_1) with PTT
c. USB (channel A_1) continuously keyed
d. LSB (channel B_1) continuously keyed
e. Two channels ISB (channels A_1 and B_1) continuously keyed
f. Four channels ISB (channels A_2, A_1, B_1, and B_2) continuously keyed, except as provided for in 3.4.12.5.4.

3.4.12.4.2 Class of emission control. The class of emission control shall provide carrier insertion levels and emission capabilities as specified in a through g:

a. AØ operation, continuous carrier insertion at rated avg power output, no modulation will be accepted
b. A_1, F_1, or F_4 operation, providing the capability of the exciter accepting these forms of modulation in the A channel
c. A_2, or A_3 operation, this function shall be interlocked with the sideband selector control to provide operation with modulation in the A_1 channel and carrier insertion 6 dB below rated PEP output
d. Carrier insertion 10 dB below rated PEP output
e. Carrier insertion 20 dB below rated PEP output
f. Carrier insertion 40 dB below rated PEP output
g. Fully suppressed carrier (at least 60 dB below rated PEP output)

3.4.12.5 Signal input. The exciter shall accept CW, FSK, and FAX audio modulating signals from the KY-655/FRT keyer unit, hereinafter referred to as the keyer, and modulation from nominal 600-ohm audio lines.

3.4.12.5.1 CW, FSK, and analog. CW, single channel FSK, and FAX (analog) keying shall be accomplished by the keyer which shall be capable of accepting CW keying at speeds up to 200 bauds, FSK up to 300 bauds, and an analog input at speeds up to 400 bauds. The electronic keyer shall be as specified in 3.4.13.

3.4.12.5.2 Keyer interface. The exciter and the keyer shall be provided with interface connectors so that when the A_1, F_1, or F_4 class of emission is selected at the exciter, the keyer is automatically placed in operation. Utilization of this requirement may be selected at the option of the installation or transmitting site personnel.

3.4.12.5.3 Modulating signals (audio). The exciter shall be capable of being modulated with voice or composite AF signals (from Navy terminal equipment) at a nominal level of 0 dBm. The nominal input level for each input shall be 0 dBm. Attenuators shall be provided for accommodation of input level from +10 dBm to -25 dBm in increments of not more than 1 dB.

3.4.12.5.3.1 VFG. Provisions shall be made in the exciter audio input channel A_1 for a VFG circuit equipped with an internal adjusted control for setting the gate sensitivity to any signal sensing level ranging from -5 dBm to -20 dBm. The output of the VFG shall be translated to dry-contact keying of the transmitter PTT control circuitry specified in 3.4.5.3.

3.4.12.5.3.1.1 VFG keying line. The dry-contact keying line from the VFG shall be capable of being switched in or out of the transmitter's internal PTT circuit.

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3.4.12.5.4 Two ISB simplex operation. The existing design of the transmitter is such that simplex operation can only be realized when using the SSB mode of transmission. The transmitter designed to the performance requirements of this specification shall have the capability for simplex operation when the two ISB mode is selected for use. A two-position internal switch shall be provided in the exciter for this purpose. One position of the switch shall be marked NORMAL and the other position shall be marked SIMPLEX 2 ISB. When the switch is placed in the NORMAL position, the transmitter shall provide key-down operation in the two ISB mode as is currently available in the GFE. When the switch is placed in the SIMPLEX 2 ISB position, the transmitter shall provide this type of operation without the necessity for any other form of conditioning. When the switch is in the SIMPLEX 2 ISB position, the transmitter shall be capable of being keyed by signals of +5 volts direct current (VDC) to +7 VDC and shall present a minimum load of 5000 ohms to the data keyline. When all of the keyline inputs are between 0 VDC and +0.5 VDC, the exciter shall be in a key-up condition. When keyline inputs are greater than +0.5 VDC, the exciter shall be held continuously in a key-down condition. Keying attack time shall be such that the transmitter RF output shall be within 1 dB of steady-state, without ringing, within 8 ms following data keyline closure (audio continuously applied). The keying decay time shall be such that the transmitter RF output shall comply with the key-up condition within 8 ms following keyline inputs between 0 VDC and +0.5 VDC. The addition of a simplex operational capability in the two ISB mode shall not degrade performance characteristics of the transmitter when operating in any mode specified herein.

3.4.12.5.4.1 Transmit-receive (T/R) relay assembly. The transmitter shall be equipped with a T/R relay assembly with a switching time less than 10 ms. The relationship between T/R relay operate and release times and the corresponding AM-6064( )/FRT keying time constants shall be such as to prevent hot switching of the transmitter RF output. The relay specified in MIL-R-83725/4 shall be used in the T/R relay assembly. A receiver antenna input receptacle shall be provided. The relay shall include provisions for grounding the receiver antenna input circuit when in the transmit condition. Cables and connectors shall be selected to permit by-passing of the relay if a failure occurs.

3.4.12.5.4.2 Receiver monitor sample. A monitor sample of the transmitter RF output shall be provided to the associated receiver's input terminals during transmissions. The amplitude of this RF output sample, measured into a 50-ohm load, shall be -40 dBm ±6 dB over the full operating frequency range of 2.0000 MHz to 29.9999 MHz. Conformance to this requirement shall be achieved when the transmitter is operated in the USB, LSB, and ISB mode with a single tone audio input of 0 dBm to the appropriate audio input terminals.

3.4.12.5.5 Audio impedance. The input impedance for each of the four voice frequency inputs shall be 600-ohms nominal. There shall be a minimum of 30 dB return loss from 370 Hz to 3040 Hz for each channel.

3.4.12.5.5.1 Electrical balance. The electrical balance shall be such that longitudinal currents are at least 40 dB below the reference level (0 dBm at 1000 Hz).

3.4.12.6 Individual subchannel gain. Each of the audio input channels shall be provided with a single front panel control which controls the amount of output power in its respective portion of the output spectrum. It shall be possible, through the use of these controls, to allow from 5 percent to 85 percent of the available output power in any one of the audio channels. Thus, it shall be possible, by using these controls, to have any combination of power division (totaling not more than 100 percent) among the four (or two) channels between the limits of 5 percent minimum and 85 percent maximum in any channel. Such controls shall be continuous and shall be calibrated between 0 percent to 70 percent in 5 percent increments, and 70 percent to 100 percent in 10 percent increments. Accuracy of calibration shall be 5 percent of the maximum value as measured in the transmitter output power.

3.4.12.7 Automatic level control. The audio inputs shall be arranged so that each input signal, at any level above 0 dBm, shall automatically and independently be attenuated in a smooth transition to the point where a +10 dBm input signal level shall not exceed +3.0 dBm to the amplifier. (The transmitter linear amplifier shall also be similarly automatically protected from high signal levels that would otherwise exceed the power limits of the transmitter and cause a shutdown.)

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3.4.12.8 Volume unit (VU) input meter. One VU meter shall be included with a provision for switching to the input of any one of the four audio circuits. The meter shall electrically follow the attenuators (see 3.4.12.5.3) and be so arranged that the meter indicates 0 dBm when the attenuator is in the 0 dBm attenuation position and a 0 dBm 1-kHz signal is applied to the exciter audio input terminal. This meter shall have performance characteristics described in IEEE 152-1953 and three percent or better accuracy.

3.4.12.9 Carrier suppression. The carrier shall be suppressed at least 60 dB below the avg power of a single-tone signal at full rated output. Means shall be provided for inserting the carrier at levels of 0 dB (no suppression), -6 dB, -10 dB, -20 dB, and -40 dB.

3.4.12.10 Carrier compression. When set for 20 dB carrier suppression, the carrier level shall not be compressed more than 1 dB at any power level from zero to full rated power output.

3.4.12.11 Carrier change under modulation. When the carrier suppression is at maximum (60 dB or more), the carrier level shall not increase more than 2.5 dB at any output level between zero and full rated output power.

3.4.12.12 Equal sideband power. When operating in the ISB mode and with equal power input to each channel, the total power output shall be divided equally among the channels within ±5 percent.

3.4.12.13 Exciter response characteristics. Exciter response characteristics shall be as specified in 3.4.12.13.1 through 3.4.12.13.4.1.

3.4.12.13.1 Four channel ISB configuration. The four channels of the exciter shall be identified in accordance with FIGURE 1.

3.4.12.13.1.1 Output channel inversion. The inversion of channels A_2 and B_2 with respect to channels A_1 and B_1 shall be accomplished by SSB modulation of subcarrier frequencies which are removed 6290 Hz above and below the output carrier frequency (f_o), respectively.

3.4.12.13.1.2 Stability. The frequency stability of channels A_2 and B_2 shall be identical with that of channels A_1 and B_1.

3.4.12.13.1.3 Inversion carriers and spurious suppression. The subcarriers used to accomplish the spectra inversion of channels A_2 and B_2 shall not degrade the specified performance requirements of the transmitter. To this end, the carriers and any of the unwanted products of such processes shall be included in the determination of intermodulation products and spurious response of the transmitter.

3.4.12.13.2 In-band response. The response of each audio channel shall be such that the output of any single modulation frequency does not vary, for a constant input level, more than 1 dB over the range of 370 Hz to 3040 Hz for channels A_1, A_2, B_1, and B_2.

3.4.12.13.3 Input and output references. The reference points for each exciter input audio response characteristic shall be 0 dBm, 1000 Hz, sinewave at the audio input terminals of each exciter channel. The audio response of each channel shall be determined by measurement in its respective portion of the modulated RF output of the exciter.

3.4.12.13.3.1 Stability of characteristics. The response of each audio channel of the exciter shall not exceed the limits specified in 3.4.12.13.2 under any allowable conditions of modulation. This requirement means that under any combination of allowable input types and levels in the other three channels, each of the channel response curves shall not exceed the limits specified in 3.4.12.12 and 3.4.12.13.2.

3.4.12.13.4 Out-of-band response. Filters shall be provided in the exciter with characteristics that will provide the out-of-band performance of the exciter for each channel as specified herein. The slope of the out-of-band response curves shall be monotonic. Conformance to the specified performance shall be achieved with the exciter installed in the transmitter and operation at any temperature within the ambient range 0°C to 50°C.

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Frequency		Attenuation
Channels B_1 and A_2	Channels A_1 and B_2	
f_o	f_o	At least -20 dB
f_o +350 Hz	f_o -250 Hz	At least -50 dB
f_o +350 Hz	f_o +350 Hz	At least -60 dB
f_o -3350 Hz	f_o +3350 Hz	At least -20 dB
f_o -3590 Hz	f_o +3590 Hz	At least -60 dB

f_o = frequency of carrier or subcarrier for each channel

3.4.12.13.4.1 Suppressed sideband attenuation. When operating in any available sideband mode, the suppressed sideband shall be attenuated at least 60 dB below the level of the unsuppressed sideband.

3.4.12.14 Envelope delay. The overall envelope delay distortion for each sideband shall not exceed that specified in a through c:

	Delay	Frequency
a.	500 microseconds (μs)	382 Hz to 3020 Hz
b.	1000 μs	370 Hz to 382 Hz
c.	1000 μs	3020 Hz to 3040 Hz

3.4.12.15 Exciter design requirements. The exciter furnished with the transmitter shall be an improved MD-777/FRT exciter. Improvements shall include the design requirements of 3.4.12.15.1 through 3.4.12.15.5.

3.4.12.15.1 Overload protection. Overload protection shall be provided where required.

3.4.12.15.2 Output amplifier module. The output amplifier module shall be capable of withstanding a direct short or open circuit on the output while operating without causing damage.

3.4.12.15.3 Transmitter gain control (TGC) module. The TGC module shall be digital in operation.

3.4.12.15.4 Synthesizer module. State-of-the-art integrated circuits shall be used in the count-down string portion of the synthesizer module.

3.4.12.15.5 Down converter module. Development of a more stable input gain control is a requirement for the down-converter module intermediate frequency (IF) amplifier.

3.4.12.16 Exciter power control. The power output level of the exciter shall be variable from 10 mW to at least 250 mW by a single control while retaining all of the performance characteristics required by this specification. The maximum output level when in the minimum position shall be no more than 10 mW. The control shall be located in back of the front panel for ready accessibility when the exciter is extended for maintenance or adjustment.

3.4.12.17 Exciter output impedance. The nominal load impedance of the exciter shall be 50-ohms resistive. Maximum VSWR at the exciter output fitting shall not exceed 1.5:1 throughout the required frequency range specified in 3.4.1. The exciter output shall be through a connector designed to mate with a standard 50-ohm Type N fitting in accordance with MIL-C-39012.

3.4.12.18 Exciter standby condition. For definition of exciter standby condition see 6.4.5.

3.4.12.18.1 Standby power. Power drain of the exciter in the standby condition shall not exceed 100 W.

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3.4.12.18.2 Transfer time and controls. It shall be possible to place the exciter in standby from full operating condition, or return it to full operating condition from standby, within 1 second, using not more than one front panel control.

3.4.12.19 Controls-indicators. The exciter shall provide, in addition to those already specified, the control-indicators specified in 3.4.12.19.1 through 3.4.12.19.6. These shall function as normal operating controls of the transmitter.

3.4.12.19.1 Amplifier OFF. The amplifier OFF control-indicator shall turn the AM-6046( )/FRT amplifier off. The indicator shall show the actual status of the transmitter.

3.4.12.19.2 Standby. The standby control-indicator shall provide a command for the transmitter to go to a standby condition. The indicator shall show the actual status of the transmitter. Positive visual indication of both standby and full operation shall be provided on the front panel of the exciter. From the appearance of these two indicated conditions, it shall be possible to deduce the third operational state: fully deactivated.

3.4.12.19.3 Operate. The operate control-indicator shall be illuminated to show transmitter operate condition. The control shall command the transmitter above standby to a full operate (that is, all voltages applied) condition.

3.4.12.19.4 Tune. The tune control-indicator shall illuminate to indicate the transmitter needs to be retuned (for example, change in frequency or class of emission). The control shall initiate a tune sequence in the transmitter and shall automatically provide an operate command.

3.4.12.19.5 Ready. The ready indicator shall illuminate to indicate the transmitter is in operate, has completed a tune sequence, and is ready for the beginning of data transmission.

3.4.12.19.6 Local and remote operation. A local-remote switch shall be provided on the exciter. In the local position the data outputs from the LCIU shall be inhibited and system control shall be executed from the front panel of the exciter. The local control condition shall be encoded by the LCIU for notification of the RCU by the monitor message reply word. Status information, with the exception of frequency, shall be contained in the monitor message reply word when the system is in the local control mode.

3.4.12.20 Transmitter status readback function. Provisions shall be made in the exciter for sensing, monitoring, and translating transmitter conditions to dry-contact control of a remote readout indicating system. The remote readout function shall be:

Modulation channel	Condition	Contacts
A_1	On	Closed
B_1	On	Closed
A_2 and B_2	On	Closed
Parameter sensed	**Condition**	**Contacts**
B+	On	Closed
RF	On	Closed

3.4.12.21 Power supply. The exciter shall operate from a single-phase power source of 115 V, ±10 percent, and 230 V, ±10 percent, 47 Hz to 63 Hz. Conversion from operation at 115 V to 230 V, or vice versa, shall be accomplished by a single toggle switch located in the exciter power supply module.

3.4.13 Electronic keyer. The electronic keyer shall be located either as an integral part of the transmitter equipment rack or shall be capable of being installed at a remote location. The electronic keyer shall be rack-mounted on a standard, Type C, 48.3-cm (19-in.) panel in accordance with EIA RS-310-C-77 and shall be capable of being connected to the exciter by a single multi-wire cable.

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3.4.13.1 Front panel controls. The keyer shall include the additional features specified in a through c:

a. A front panel switch shall allow operator selection of teletypewriter (TTY), CW, or FAX modes of operation.

b. A direct reading 10-turn TTY deviation control shall allow operator selection of deviations from 1 Hz to 999 Hz.

c. A front panel function test switch shall be provided to allow operator selection of TTY mark, TTY space, CW, FAX white, and FAX black for test purposes.

3.4.13.2 Interface with exciter. The keyer and the exciter shall be provided with interfaces that shall allow the keyer to be connected to and controlled by the exciter. When so connected, selection of the A_1, F_1 or F_4 class of emission shall automatically place the keyer in operation. Terminal strips with appropriate connections shall be available within the keyer so that the utilization of this requirement may be selected by the installation or transmitter site personnel.

3.4.13.2.1 Automatic line transfer. The electronic keyer shall provide a means for automatically connecting the output to channel A_1 whenever A_1, F_1, or F_4 is selected, and shall select an external audio line which is terminated into channel A_1 of the exciter.

3.4.13.3 Keying speeds. The electronic keyer shall be capable of operation with CW keying up to 200 bauds, FSK (TTY) up to 300 bauds, and analog inputs (FAX) at speeds up to 400 bauds.

3.4.13.4 Keyer output frequencies. The separation between mark and space tones provided by the keyer for frequency-shift TTY operation shall be continuously adjustable from 1 Hz to 999 Hz. Accuracy of all output tones shall be within ±1 Hz or ±1 percent, whichever is greater. The stability of all tones shall be within 2 Hz per day. The keyer shall provide a 1000-Hz tone for CW ON-OFF keying. In addition, the frequency-shift tones for teletype and FAX operation provided by the keyer shall be selectable about the center frequencies specified in a through d by means of a four-position switch:

a. 2500 Hz
b. 2000 Hz
c. 1900 Hz
d. Internally adjustable for 400 Hz to 3000 Hz

3.4.13.5 Keyer input signals. The keying sense shall be front panel selectable. The electronic keyer shall accept the types of signal input specified in a through g:

a. 20 mA neutral (external battery)
b. 60 mA neutral (external battery)
c. 50 V neutral (external battery)
d. 100 V neutral (external battery)
e. Dry relay contacts (internal battery)
f. Keyer shall accept a facsimile signal input which varies linearly between 1 V and 10 VDC and shall deliver a frequency-shift tone which varies linearly from 400 Hz above to 400 Hz below the selectable rest frequency.
g. Keyer shall also accept low level (±6 V) polar signals with input sensitivity of 0.5 V, positive mark, external battery, with no internal ground.

3.4.13.6 Input impedance. The keyer shall have the input impedance specified in a through g for input signals specified in the corresponding subparagraphs of 3.4.13.5:

a. 150 ohms, or less
b. 150 ohms, or less
c. 47,000 ohms
d. 100,000 ohms
e. 47,000 ohms
f. 47,000 ohms minimum
g. 100,000 ohms, or greater

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3.4.13.7 Keyer power supply. The keyer shall operate from single-phase power sources of 115 V, ±10 percent, and 230 V, ±10 percent, 47 Hz to 63 Hz. Conversion from operation at 115 V to 230 V, or vice versa, shall be accomplished by a single toggle switch located behind the keyer front panel and protected by a guard to prevent inadvertent switching.

3.4.13.8 CW mode. The keyer shall be designed so that CW operation is possible using 60 mA, 20 mA, 100 V, ±6 V, polar or dry contact input signals.

3.4.13.9 Direct current (DC) isolation. The keyer shall be designed so that the DC input loop in all modes of operation shall be isolated from ground.

3.4.14 Linear Amplifier, AM-6046( )/FRT and associated power supply. Linear Amplifier AM-6046( )/FRT and associated power supply shall be as specified in 3.4.14.1 through 3.4.14.17.4.

3.4.14.1 Frequency range and tuning. The frequency range of the HF linear amplifier, hereinafter referred to as the amplifier, shall be continuous from 2.0000 MHz to 29.9999 MHz plus or minus half the transmitted intelligence bandwidth. Normally, the amplifier shall accept coded channel information from the exciter and automatically select the proper operating band from those specified in 3.4.14.1.1. The transmitter shall be capable of being automatically tuned on any frequency from 2.0000 MHz to 29.9999 MHz in 10 seconds or less when the output is terminated in a load producing a VSWR up to 4.0:1. The operating band for the amplifier shall be manually selectable by a front panel control when required.

3.4.14.1.1 Primary frequency bands. The amplifier shall include 19 primary frequency bands as specified in a through c:

a. Four bands of 0.5 MHz bandwidth from 2.0 MHz to 4.0 MHz
b. Four bands of 1.0 MHz bandwidth from 4.0 MHz to 8.0 MHz
c. Eleven bands of 2.0 MHz bandwidth from 8.0 MHz to 30.0 MHz

3.4.14.1.2 Tuning method. The PA shall be tuned without RF energy in the 19 frequency bands specified in 3.4.14.1.1. Frequency information shall be conveyed by a five-wire binary code supplied by the exciter and repeated by an interface box. Frequency tuning of the amplifier shall be automatically accomplished when a frequency is selected and the tune control on the exciter is depressed.

3.4.14.2 Power output. The amplifier shall amplify any emission presented to its input from the exciter, in a linear manner, within the limits of a 25-kHz maximum intelligence bandwidth. The power output delivered to a 50-ohm unbalanced nonreactive load over the frequency range of 2.0000 MHz through 29.9999 MHz plus or minus half the transmitted intelligence bandwidth shall be at least as specified in 3.4.14.2.1 when driven by the input power specified for each mode of operation. The amplifier shall have provisions for automatically regulating the power output with an antenna system having VSWR greater than 1:1 such that no damage will be caused to the transmitter. This requirement shall be accomplished by reduction of the output power level with increasing VSWR and by cathode current limiting in accordance with 3.4.14.2.5. The forward output power level shall be not less than 400 W avg in the A_1 and F_1 modes with an antenna system having a VSWR of 4:1.

3.4.14.2.1 Power outputs for specified inputs. The PA outputs for specified inputs are:

Input	Output
SSB (A3J): 200 mW (PEP two-tone)	At least 1 kW (PEP two-tone)
CW or FSK (A_1 or F_1): 200 mW avg	At least 1 kW avg (single-tone)

3.4.14.2.2 Avg noise power output rating. The minimum avg noise power output level of the trasmitter shall be 200 W when three of the four nominal 3-kHz channels are modulated at the appropriate audio level. Under this loading condition the intermodulation levels of 3.4.7.1 shall be achieved.

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3.4.14.2.3 Power level control. A power output level control shall be located on the amplifier front panel. The power level control shall be protected by a cover and shall permit continuous adjustment of the output power to any desired level between the limits of the amplifier's maximum power rating and 25 percent of that rating. The power level control shall operate independently from any other control and shall not degrade any of the operating parameters of the amplifier or exciter other than output power.

3.4.14.2.4 Antenna Coupler Group, AN/URA-38 compatibility. The transmitter shall interface with the Antenna Coupler Group, AN/URA-38 specified in MIL-C-28806 but shall be capable of operation with or without the AN/URA-38 equipment. Tests shall be conducted to ensure compatibility of the transmitter and the AN/URA-38.

3.14.2.5 Cathode current limiting. The PA shall protect the final tubes and power supply against excessive current by automatically reducing drive to limit the cathode current to no more than 750 mA.

3.4.14.3 Information bandwidth. With an RF input signal of constant level driving the amplifier at any frequency between 2.0000 MHz and 29.9999 MHz and with the amplifier terminated into a 50-ohm resistive load, it shall be possible to change the driving frequency by ±12.5 kHz without causing more than 1 dB variation in the amplifier output power level.

3.4.14.4 Regeneration. There shall be no evidence of unintended oscillation or instability under any condition of operation of the amplifier when employing tubes with the maximum transconductance specified for the type tubes used with a supply voltage up to 10 percent higher than the nominal primary voltage.

3.4.14.5 Harmonic output. Output of the second order harmonic shall be at least 45 dB below the power level of the fundamental signal at all power levels up to and including rated output. Higher order harmonic products shall be at least 55 dB below the power level of the fundamental signal at all power levels up to and including rated output.

3.4.14.6 Intermodulation, spurious emissions, and hum. Intermodulation product and spurious emissions produced in the amplifier shall be limited to levels consistent with the overall requirements for the transmitter, as specified in 3.4.6 for spurious emissions, and in 3.4.7 for intermodulation distortion. In-band hum products shall be not less than 43 dB below a 1-kW single-tone output reference level at any frequency within the range of the selected amplifier frequency band.

3.4.14.7 TGC. An electronic TGC circuit shall be provided in the amplifier to generate a control voltage which, when interfaced with the exciter, shall automatically adjust the transmitter power output during the tuning cycle so that power output during operation is set to rated avg power within a ±1.0 dB limit. Adjustable controls associated with this circuit shall be equipped with locking devices.

3.4.14.8 Peak power control (PPC). A PPC circuit shall be provided in the amplifier to generate a control voltage which, when interfaced with the exciter, shall prevent the peak power output of the transmitter from exceeding the limits of this specification on a preset level. Adjustable level controls associated with this circuit shall be equipped with locking devices and shall be located inside the amplifier. The PPC time constant shall be between 1 ms and 2 ms attack time with approximate critical damping and between 60 ms and 100 ms decay time. The circuitry shall be such that special readjustment of the controls is not required when changing operating modes, for example, changing from A3J to A3B.

3.4.14.9 Keying transient. Provision shall be made for eliminating any key-down transients originating in the transmitter.

3.4.14.10 Meters. The RF output power meter on the tuner emulator panel shall provide scales and switching for measurement of forward and reflected power as specified in a and b:

	Forward	Reflected
a.	0 to 1500 W PEP	0 to 1500 W PEP 0 to 1500 W avg
b.	0 to 1500 W avg	0 to 150 W PEP 0 to 150 W avg

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The scales shall be calibrated logarithmically with the minimum scale calibration mark of 100 W for the 1500-W scale and 10 W for the 150-W scale. One switch shall be of the two-position rotary type and shall select either PEP or avg power indications. A spring-loaded, three-position switch shall select the forward and reflected power indications and shall be so arranged that forward power is indicated in the center position. The spring-loading shall provide automatic return of the switch to the forward power (center) position. Calibration of the RF output power meter shall be accurate two within 10 percent over the 2.0000 MHz through 29.9999 MHz frequency range, for single-tone, two-tone, 16-tone conventional tone library used with teletype terminal equipment, and noise loaded signals. Metering of the amplifier input from the exciter shall be provided on the amplifier's multipurpose meter. The input shall be measured on a 0 mW through 150 mW scale.

3.4.14.11 Amplifier input impedance. Throughout the frequency range of 2.0000 MHz through 29.9999 MHz, the amplifier input circuit impedance shall be nominal 50-ohms resistive with reactive components reduced to values which will cause a VSWR no higher than 1.5:1 at the exciter output.

3.4.14.12 VSWR output monitor. A VSWR monitor circuit shall be provided for automatically regulating the amplifier power output to a lower level, without interruption of traffic, when the reflected power is equivalent to that produced by a VSWR of between 2.5:1 and 4.0:1 at rated output power.

3.4.14.13 Efficiency. The overall efficiency of the amplifier shall be not less than 20 percent determined from the 50-Hz or 60-Hz power input with the transmitter loaded with a two-tone signal to 1-kW PEP output.

3.4.14.14 Amplifier power supply. The power supply shall provide all the necessary operating voltages. Short circuits applied to any power supply output shall cause no damage to the power supply. Indicating fuseholders shall provide a visible indication of status. Solid-state silicon devices shall be used in lieu of vacuum tubes.

3.4.14.15 Amplifier standby. The amplifier standby condition is identical to the exciter standby condition, with the addition specified herein. All components and circuitry in the amplifier requiring more than 30 seconds to reach full operating condition shall remain activated during standby. All circuitry remaining active during the standby condition shall be so protected that the cabinet may be opened for maintenance without violating any personnel safety requirements (see 3.2.7).

3.4.14.16 RF sampling. Individual RF output sampling receptacles shall be provided to enable the monitoring of RF input and output spectra of the amplifier. Sampling shall be provided at levels specified in 3.4.14.16.1 through 3.4.14.16.4 based on modulation with a standard two-tone signal and the transmitter delivering full rated 1-kW output into a 50-ohm load. The intent of the sampling receptacles is to allow monitoring of the transmitter signal within the normally transmitted bandwidth of 12.6 kHz.

3.4.14.16.1 Rear panel input connector. One coaxial connector of 50-ohms impedance shall be mounted on the rear of the tuner emulator panel and shall be labeled INPUT MON. The connector shall be designed to mate with a standard 50-ohm, Type BNC fitting in accordance with MIL-C-39012 and connected internally to monitor the amplifier input. The 1 Vrms level shall be based on 100 mW avg power input.

3.4.14.16.2 Front panel input connector. One coaxial connector of 50-ohms impedance shall be mounted on the front of the tuner emulator panel and shall be labeled INPUT MON. The connector shall be designed to mate with a standard 50-ohm, Type BNC fitting in accordance with MIL-C-39012 and connected internally to monitor the amplifier input. The 1 Vrms level shall be based on 100 mW avg power input.

3.4.14.16.3 Rear panel output connector. One coaxial connector of 50-ohms impedance shall be mounted on the rear of the tuner emulator panel and shall be labeled OUTPUT MON. The connector used for this output shall be designed to mate with a standard 50-ohm, Type BNC fitting in accordance with MIL-C-39012 and connected internally to monitor the amplifier output. The 0.2 Vrms minimum level shall be based on 1000 W avg power output.

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3.4.14.16.4 Front panel output connector. One coaxial connector of 50-ohms impedance shall be mounted on the front of the tuner emulator panel and shall be labeled OUTPUT MON. The connector shall be designed to mate with a standard 50-ohm, Type BNC fitting in accordance with MIL-C-39012 and connected internally to monitor the amplifier output. The 0.2 Vrms minimum level shall be based on 1000 W avg power output.

3.4.14.17 Amplifier design changes. Amplifier design changes shall be as specified in 3.4.14.17.1 through 3.4.14.17.4.

3.4.14.17.1 PA screen supply protection. The PA screen power supply shall be protected from damage due to a plate-to-screen short.

3.4.14.17.2 Output coupling capacitor C4. The output coupling capacitor C4 shall have a rating of not less than 4000 V at 135°C and 12 amperes at 30 MHz.

3.4.14.17.3 Manufacturer's part numbers. The design of the transmitter shall utilize modules, parts, and assemblies with equivalent part numbers to Harris Corporation, RF Communications Division as specified in a through i:

a.	Final transformer assembly:	10051-3700
b.	Auxiliary frequency generator:	10050-1200
c.	1.75/113.75 MHz generator:	10050-1100
d.	Synthesizer:	10050-2000
e.	Down-converter:	10050-1400
f.	TGC:	10050-1300
g.	Output amplifier:	10050-1600
h.	Power supply:	10050-3000
i.	Sideband selector switch:	10050-0100

3.4.14.17.4 High-voltage rectifiers. Type JANTX 1N5554-1000 V, 3-ampere, 100-ampere surge, high-voltage rectifiers in accordance with MIL-S-19500/420 shall be used.

3.4.15 Remote control system. The transmitter shall be capable of being remotely controlled by the Remote Control System, AN/FSQ-98(V) which conforms to the requirements specified herein and which interfaces with the transmitter through the exciter. The extent of the remote control shall be limited to those functions specified in TABLE I, TABLE II, and TABLE III.

TABLE I. Exciter connector J7 MS3116E20-39P.

Terminal	Function	
A		
B		
C	100 Hz	
D		
E		
F	1 kHz	
G		
H		
J		
K	10 kHz	Frequency selection (see TABLE III)
L		
M		
N		
P	100 kHz	
R		
S		
T		
U	1 MHz	
V		
W		
X	10 MHz	
Y		
Z	VCO A	

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TABLE I. Exciter connector J7 MS3116E20-39P. (continued)

Terminal	Function	
a	VCO B	Class of emission selection
b	AO	
c	A2, A3H	
d	A1, F, F4	
e	SSB -10 dB carrier	
f	SSB -20 dB carrier	
g	SSB -40 dB carrier	
h	SSB -infinity carrier	
i	Standby command	
j	Operate command	
k	Ground	
m	Ground	
n	PTT command	

TABLE II. Exciter connector J8 MS3116E20-39PW.

Terminal	Function
A	USB select control
B	LSB select control
C	ISB (2) select control
D	ISB (4) select control
G	AO class of emission readback
H	A2, A3H class of emission readback
J	A1, F1, F4 class of emission readback
K	SSB -10 class of emission readback
L	SSB -20 class of emission readback
M	SSB -40 class of emission readback
N	SSB infinity class of emission readback
P	Operate readback
R	Standby readback
S	USB readback
T	LSB readback
U	ISB (2) readback
V	ISB (4) readback
Y	Ready readback (common)
Z	Ready readback
a	Fault readback
b	Fault readback (common)
c	Local status
d	Transmitter not available readback (common)
e	Tune command
f	Tune activate common (ground)
g	+5 VDC
h	PTT readback
i	Transmitter fail
j	VFG readback
k	Standby-operate readback (common)
m	Sideband select (common)
n	Sideband readback (common)
q	Class of emission readback (common)
v	VFG command

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TABLE III. Frequency selection tables.

100-Hz digit selection				
DIGIT	J7-A	J7-B	J7-C	J7-D
0	0	0	0	1
1	1	0	0	1
2	0	0	1	1
3	1	0	1	1
4	0	1	1	0
5	1	1	1	0
6	0	1	0	0
7	1	1	0	0
8	0	0	0	0
9	1	0	0	0
1-kHz digit selection				
DIGIT	J7-E	J7-F	J7-G	J7-H
0	1	0	0	1
1	0	0	0	1
2	1	0	1	1
3	0	0	1	1
4	1	1	1	0
5	0	1	1	0
6	1	1	0	0
7	0	1	0	0
8	1	0	0	0
9	0	0	0	0
10-kHz digit selection				
DIGIT	J7-J	J7-K	J7-L	J7-M
0	0	0	1	1
1	1	1	0	1
2	0	1	0	1
3	1	0	0	1
4	0	0	0	1
5	1	1	1	0
6	0	1	1	0
7	1	0	1	0
8	0	0	1	0
9	1	1	0	0
100-kHz digit selection				
DIGIT	J7-N	J7-P	J7-R	J7-S
0	0	0	0	1
1	1	0	0	1
2	0	0	1	1
3	1	0	1	1
4	0	1	1	0
5	1	1	1	0
6	0	1	0	0
7	1	1	0	0
8	0	0	0	0
9	1	0	0	0

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TABLE III. Frequency selection tables. (continued)

1-MHz digit selection				
DIGIT	J7-T	J7-U	J7-V	J7-W
0	1	1	0	0
1	0	1	0	0
2	1	0	0	0
3	0	0	0	0
4	1	0	0	1
5	0	0	0	1
6	1	0	1	1
7	0	0	1	1
8	1	1	1	0
9	0	1	1	0

10-MHz digit selection		
DIGIT	J7-X	J7-Y
0	1	1
1	0	1
2	1	0

VCO selection		
Frequency range (MHz)	J7-Z (VCO A)	J7-a (VCO B)
2.0000 to 11.9999	0	0
12.0000 to 21.9999	0	1
22.0000 to 29.9999	1	0

3.4.15.1 Primary power. All units of the remote control system shall operate from a 115 V ±10 percent, single-phase, 47-Hz to 63-Hz primary power source.

3.4.15.2 Remote control interface. All interfaces between the remote control system and the transmitter shall be via the remote system's LCIU. This LCIU shall be located in a space provided in the transmitter cabinet as shown in NAVELEX DWG 0100315D. All control signals necessary to satisfy system operation shall be exchanged between the LCIU and the exciter.

3.4.15.3 LCIU location. The cabinet shall provide a space capable of accepting the LCIU. The LCIU shall be contained in an enclosed cabinet suitable for mounting in a standard 48.3-cm (19 in.) equipment rack as specified in EIA RS-310-C-77. LCIU dimensions shall not exceed 18 cm (7 in.) high, 45.7 cm (18 in.) deep, and 43.5 cm (17.125 in.) wide, exclusive of the front panel mounting. The LCIU shall be located in the cabinet as shown in NAVELEX DWG 0100315D. Primary power for the LCIU shall be available on the standard multiple outlet strip specified in 3.4.11.1.

3.4.15.4 Transmitter control. All information received by the LCIU from the RCU shall be converted to parallel data for application to the exciter control lines. The remote controlled signaling shall be decoded as necessary in the LCIU to achieve compatibility with the exciter control lines and functions. Inversely, monitor information from the exciter shall be encoded by the LCIU for transmission to the RCU.

3.4.15.4.1 Transmitter control and monitor connections. Actual control of the transmitter and monitoring of the transmitter state of operation when in a remote control mode shall be via two connectors located on the exciter rear panel. Cables shall be provided with the LCIU to mate with these connectors. The connector types and pin functions are specified in TABLE I, TABLE II, and TABLE III.

3.4.15.4.2 LCIU and exciter interface signal levels. LCIU and exciter interface signal levels shall be as specified in 3.4.15.4.2.1 and 3.4.15.4.2.2.

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3.4.15.4.2.1 LCIU to exciter. Unselected (Logical 0) state = 0.0 V to 0.5 V (current sinking capability of 1.5 mA); selected (Logical 1) state = 1.5 V to 5.0 V (maximum current = 100 microamperes (μA). The exciter control input circuitry shall provide a current sink for the LCIU unit logic control circuitry. The 5.0 VDC for the interface circuitry contained in the LCIU shall be provided by the exciter to minimize ground currents between units. A maximum current of 10 mA shall not be exceeded.

3.4.15.4.2.2 Exciter to LCIU. Unselected (Logical 0) state = closed circuit to LCIU common line. When the status of the exciter is changed by the remote control system, readback lines associated with the changed parameters may provide invalid data for a period of up to 1.5 seconds.

3.5 Environmental conditions. The transmitter shall conform to the performance requirements of this specification under the conditions specified in a and b:

a. Operating:
   1. Ambient temperature: 0°C to 50°C, at sea level
   2. Humidity: 0 percent to 95 percent relative humidity (RH)
   3. Elevation: Sea level to 3000 m (approximately 10,000 ft), 20°C at 50 percent RH
b. Nonoperating:
   1. Ambient temperature: -62°C to +71°C
   2. Elevation: Sea level to 15,000 m (approximately 50,000 ft)

3.5.1 Thermal design. The thermal design of the transmitters shall be in accordance with the requirements of the Thermal design paragraph (and subparagraphs) of MIL-E-16400, except Thermal instrumentation, and that inlet-to-outlet temperature differential shall not exceed 50°C.

3.5.2 Dust. The transmitter shall be designed and constructed to withstand the dust test of MIL-STD-810, Method 510.2, Procedure I without performance degradations or malfunctions.

3.5.3 Shock. The transmitter shall be designed and constructed to withstand the shock imposed when tested in accordance with MIL-STD-810, Method 516.3, Procedure I, and the incline impact test listed in the Rough handling tests table of MIL-P-116, without performance degradations or malfunctions.

3.6 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.6.1 Selection of parts. The equipment furnished in accordance with this specification shall employ standard parts or approved nonstandard parts.

3.6.2 Nonstandard parts. All nonstandard parts shall be approved and documented in accordance with the requirements of MIL-E-16400 prior to the submission of the first article sample (FAS) for Government inspection and approval. The FAS shall incorporate standard parts and approved nonstandard parts prior to conducting the FAS inspection.

3.6.3 Electron tubes and solid-state devices. Solid-state devices shall be employed instead of electron tubes to the maximum extent consistent with the performance requirements of this specification and for general specification application. The number of different types of solid-state devices and electron tubes shall be kept to a minimum. All electron tubes and solid-state devices shall be operated well below their maximum specified rating to ensure conformance to the mean-time-between-failure (MTBF) requirement of 3.8. Where applicable, overload protection shall be provided. The contractor shall be guided by MIL-E-16400 in the selection of solid-state devices. The transmitters produced in accordance with this specification shall not contain germanium solid-state devices. All diodes and transistors used in the transmitter shall be silicon.

3.6.3.1 Solid-state devices in the exciter. The exciter shall employ 100 percent solid-state devices.

3.6.4 Variable capacitors. Variable capacitors used for the higher power tuning elements (servo controlled) shall be vacuum, gas filled ceramic, or glass type and shall be rated to withstand at least 1.5 times the maximum voltage and current that might be encountered under the worst-case of operating conditions implied by the provisions specified herein.

3.6.5 Resistors. Fixed composition resistors shall be ±10 percent or better tolerance. Resistors having a tolerance of less than ±10 percent shall be limited to cases where circuit use makes them mandatory to attain the performance requirements of this specification.

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3.6.6 Encapsulation. Plastic encapsulated transistors or microelectronic devices shall not be used in the design of the equipments covered by this specification unless specifically approved, in writing, by the procuring activity.

3.6.7 Electrical contacts. Friction or pressure contacts, when used, shall be designed to prevent erratic operation caused by the effects of vibration and temperature. Whenever practicable, electrical switch and relay contacts shall be of the self-cleaning type.

3.7 Support maintenance test procedure verification. Support maintenance test procedure verification shall be as specified in 3.7.1 through 3.7.3.

3.7.1 Support maintenance. The support maintenance test procedure verification shall be performed by a technician selected by the Government from contractor or Government personnel.

3.7.2 Test equipment. Test equipment used to verify the support maintenance test procedures shall be selected from MIL-STD-1364. Test equipment not specified in MIL-STD-1364 may be used in the support maintenance test procedure verification only if the contractor submits a written request to the procuring activity and approval is granted (see 6.2).

3.7.3 Verification. The support maintenance test procedure verification shall be performed on one of the FAS.

3.8 Quantitative reliability requirements. The upper test MTBF (θ_0 as defined in MIL-STD-781) of the transmitter shall be 800 hours. The lower test MTBF (θ_1 as defined in MIL-STD-781) shall be 400 hours.

3.9 Quantitative maintainability requirements. The transmitter shall have a corrective-maintenance-time (M_{ct}) of 1.0 hour and a maximum-corrective-maintenance-time of 1.5 hours at a 90 percent confidence level when repair is accomplished at the organizational level of maintenance by the replacement of modules and chassis-mounted components (this includes electronic, electrical, electromechanical, and mechanical parts). The M_{ct} includes localization, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (PC card) for which repair is accomplished by the replacement of piece parts.

3.10 Workmanship. Workmanship shall be as specified in 3.10.1 and 3.10.2.

3.10.1 General workmanship. The workmanship and soldering requirements shall be in accordance with the Workmanship and Soldering paragraphs of MIL-E-16400.

3.10.2 Workmanship screen. All transmitters produced in accordance with this specification shall withstand a defect detection vibration screen (see 4.8.2.1) of random type vibration at $0.02g^2/Hz$ ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.8.2.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

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TABLE IV. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				In-process inspection	Production inspection
Transmitter:					
Ventilation and cooling	3.2.4	4.5.5	X		
Acoustic noise	3.2.5	4.5.6	X		
Insulating compounds	3.2.6	4.5.5	X		
Accessibility	3.2.3	4.5.5	X		
Protection of internal components	3.2.8	4.5.5	X		
Internal transient voltage protection	3.2.8.4	4.5.6	X		
Power supply protection	3.2.8.5	4.5.6	X		
Transmitter protection	3.2.8.1	4.5.2	X		
PC assembly protection	3.2.8.6	4.5.5	X		
Test points	3.2.9	4.5.6	X		
External circuit fault protection	3.2.8.2	4.5.2	X		
Test cords and module extenders	3.2.9.2	4.5.5	X		
Meters and indicators	3.2.10	4.5.5	X		
Multi-test meters	3.2.10.1	4.5.5	X		
Elapsed-time meters	3.2.10.2	4.5.5	X		
Tune cycle counter	3.2.10.3	4.5.5	X		
Controls	3.2.11	4.5.5	X		
Marking	3.2.11.1	4.5.5	X		
Manual override	3.2.11.2	4.5.5	X		
Interchangeability	3.2.12	4.5.6	X		
Reliability	3.8	4.6	X		
Safety to personnel	3.2.7	4.5.2, 4.5.6	X		
Shock	3.3.1, 3.5.3	4.5.6	X		
Dust	3.5.2	4.5.6	X		
Environmental conditions	3.5	4.5.1	X		
Thermal design	3.5.1	4.5.1	X		
Construction					
Side panel removal	3.3.1.1	4.5.5	X		
Cabinet door and fasteners	3.3.1.2	4.5.5	X		
Degree of enclosure	3.3.2	4.5.2	X		
Mountings	3.3.3	4.5.6	X		
Modular construction	3.3.4	4.5.5	X		
Frequency selection	3.4.3	4.5.6	X		
Dry contact keying	3.4.5.3	4.5.2	X		
Output impedance	3.4.8	4.5.6	X		
Power supply	3.4.11	4.5.6	X		
Exciter, keyer, and LCIU power distribution	3.4.11.1	4.5.6	X		
VSWR protection	3.2.8.3	4.5.6	X		X
Test jacks	3.2.9.1	4.5.6	X		X
Frequency range	3.4.1	4.5.6	X		X
Transmitter dimensions	3.2.1	4.5.6	X		X
Interchangeability	3.2.12	4.5.6	X		X
Frequency stability	3.4.2	4.5.6	X		X
Electrical noise	3.4.5	4.5.6	X		X
Key-up	3.4.5.1	4.5.6	X		X
Intelligence loss	3.4.5.2	4.5.6	X		X
Safety to personnel (leakage current)	3.2.7	4.5.4	X		X
In-band intermodulation distortion when white noise loaded	3.4.7.1	4.5.6	X		X
Two-tone performance	3.4.7.2	4.5.6	X		X
In-band spurious emissions and hum products	3.4.7.3	4.5.6	X		X

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TABLE IV. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				In-process inspection	Production inspection
Exciter:					
Internal frequency standard	3.4.12.2.1	4.5.6	X		
Rear panel connector	3.4.12.2.1.1	4.5.5	X		
Front panel connector	3.4.12.2.1.2	4.5.5	X		
External frequency standard	3.4.12.2.3	4.5.5	X		
Failure alarm	3.4.12.2.6	4.5.5	X		
Calibration controls	3.4.12.2.7	4.5.6	X		
Monitoring output	3.4.12.2.7.1	4.5.2	X		
Frequency controls	3.4.12.3	4.5.5	X		
Number and function of controls	3.4.12.3.1	4.5.5	X		
Operating modes	3.4.12.4	4.5.5	X		
Sideband selector control	3.4.12.4.1	4.5.5	X		
Signal input	3.4.12.5	4.5.5	X		
Keyer interface	3.4.12.5.2	4.5.5	X		
T/R relay assembly	3.4.12.5.4.1	4.5.6	X		
Receiver monitor sample	3.4.12.5.4.2	4.5.6	X		X
Audio impedance	3.4.12.5.5	4.5.6	X		
Electrical balance	3.4.12.5.5.1	4.5.6	X		
Four channel ISB configuration	3.4.12.13.1	4.5.5	X		
Output channel inversion	3.4.12.13.1.1	4.5.5	X		
Stability	3.4.12.13.1.2	4.5.2	X		
Inversion carriers and spurious suppression	3.4.12.13.1.3	4.5.5	X		
Overload protection	3.4.12.15.1	4.5.2	X		
Output amplifier module	3.4.12.15.2	4.5.6	X		
TGC module	3.4.12.15.3	4.5.5	X		
Synthesizer module	3.4.12.15.4	4.5.5	X		
Down-converter module	3.4.12.15.5	4.5.2	X		
Controls-indicators:					
Amplifier OFF	3.4.12.19.1	4.5.5	X		
Standby	3.4.12.19.2	4.5.5	X		
Operate	3.4.12.19.3	4.5.5	X		
Tune	3.4.12.19.4	4.5.5	X		
Ready	3.4.12.19.5	4.5.5	X		
Local and remote operation	3.4.12.19.6	4.5.5	X		
Transmitter status readback function	3.4.12.20	4.5.5	X		
Power supply	3.4.12.21	4.5.6	X		X
Exciter characteristics	3.4.12.1	4.5.6	X		X
Phase-locking	3.4.12.2.4	4.5.5	X		X
Automatic switching	3.4.12.2.5	4.5.6	X		X
Calibration controls	3.4.12.2.7	4.5.6	X		X
Class of emission control	3.4.12.4.2	4.5.6	X		X
CW, FSK, and analog	3.4.12.5.1	4.5.6	X		X
Modulating signals (audio)	3.4.12.5.3	4.5.6	X		X
VFG	3.4.12.5.3.1	4.5.6	X		X
VFG keying line	3.4.12.5.3.1.1	4.5.6	X		X
Two ISB simplex operation	3.4.12.5.4	4.5.6	X		X
Individual subchannel gain	3.4.12.6	4.5.6	X		X
Automatic level control	3.4.12.7	4.5.6	X		X
VU input meter	3.4.12.8	4.5.5	X		X
Carrier suppression	3.4.12.9	4.5.6	X		X
Carrier compression	3.4.12.10	4.5.6	X		X
Carrier change under modulation	3.4.12.11	4.5.6	X		X
Equal sideband power	3.4.12.12	4.5.6	X		X
In-band response	3.4.12.13.2	4.5.6	X		X
Out-of-band response	3.4.12.13.4	4.5.6	X		X
Suppressed side band attenuation	3.4.12.13.4.1	4.5.6	X		X

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TABLE IV. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				In-process inspection	Production inspection
Exciter power control	3.4.12.16	4.5.6	X		X
Standby power	3.4.12.18.1	4.5.6	X		X
Aging	3.4.12.2.2.1	4.5.6	X		X
Envelope delay	3.4.12.14	4.5.6	X		X
Exciter output impedance	3.4.12.17	4.5.6	X		X
Transfer time and controls	3.4.12.18.2	4.5.6	X		X
KY-655:					
Electronic keyer	3.4.13	4.5.5	X		
Front panel controls	3.4.13.1	4.5.5	X		
Interface with exciter	3.4.13.2	4.5.5	X		
Automatic line transfer	3.4.13.2.1	4.5.5	X		
Input impedance	3.4.13.6	4.5.6	X		
Keyer power supply	3.4.13.7	4.5.6	X		
CW mode	3.4.13.8	4.5.5	X		
DC isolation	3.4.13.9	4.5.5	X		
Keying speeds	3.4.13.3	4.5.6	X		X
Keyer output frequencies	3.4.13.4	4.5.6	X		X
Keyer input signals	3.4.13.5	4.5.6	X		X
AM-6046( )/FRT:					
Primary frequency bands	3.4.14.1.1	4.5.5	X		
Antenna Coupler Group, AN/URA-38 compatibility	3.4.14.2.4	4.5.6	X		
Keying transient	3.4.14.9	4.5.5	X		
Amplifier power supply	3.4.14.14	4.5.5	X		
Amplifier standby	3.4.14.15	4.5.5	X		
RF sampling	3.4.14.16	4.5.6	X		
Amplifier design changes	3.4.14.17	4.5.5	X		
Frequency range and tuning	3.4.14.1	4.5.6	X		X
Tuning method	3.4.14.1.2	4.5.6	X		X
Power output	3.4.14.2	4.5.6	X		X
Power outputs for specified inputs	3.4.14.2.1	4.5.6	X		X
Avg noise power output rating	3.4.14.2.2	4.5.6	X		X
Power level control	3.4.14.2.3	4.5.6	X		X
Regeneration	3.4.14.4	4.5.6	X		X
Intermodulation, spurious emissions, and hum	3.4.14.6	4.5.6	X		X
PPC	3.4.14.8	4.5.6	X		X
Meters	3.4.14.10	4.5.6	X		X
Efficiency	3.4.14.13	4.5.6	X		X
Remote control system:					
Primary power	3.4.15.1	4.5.5	X		
Remote control interface	3.4.15.2	4.5.5	X		
LCIU location	3.4.15.3	4.5.5	X		
Transmitter control	3.4.15.4	4.5.5	X		
Transmitter control and monitor connections	3.4.15.4.1	4.5.5	X		
LCIU and exciter interface signal levels:					
LCIU to exciter	3.4.15.4.2.1	4.5.5	X		
Exciter to LCIU	3.4.15.4.2.2	4.5.5	X		
General workmanship	3.10.1	4.8.1	X	X	
Workmanship screen:					
Random vibration	3.10.2	4.8.2.1		X	
Temperature cycling	3.10.2	4.8.2.2		X	
Retractable cable harness	3.2.2	4.5.5	X	X	
Finish	3.3.5	4.5.5	X	X	
Cables	3.2.2.1	4.5.3	X	X	

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4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. In-process inspection (see 4.4.1)
   2. Production inspection (see 4.4.2)
   3. Periodic inspection (see 4.4.3)
c. Inspection of preparation for delivery (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV and the tests specified in 4.4.3. FAS tests shall be conducted with the Government-furnished AN/FSQ-98(V) installed and functioning as designed.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.3.3.

4.4.1 In-process inspection. Each transmitter produced in accordance with this specification shall be subjected to the inspection specified in TABLE IV.

4.4.2 Production inspection. Production inspection, including sampling, shall conform to TABLE IV and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production inspection shall be performed on transmitters that have completed the in-process inspection. Lots shall be physically accumulated and identified prior to random selection of the sample. The Government shall make the random sample selection. Sample transmitters shall satisfactorily conform to the production inspection requirements prior to releasing the lot for shipment. The acceptable quality level shall be 6.5 percent for each attribute.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may rework the rejected lot to correct the defective transmitter and reinspect the lot using tightened inspection. As a minimum, the rework by the contractor shall consist of inspecting each transmitter in the rejected lot for the failed attribute prior to reinspection using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Periodic inspection. Periodic inspection shall be as specified in TABLE V and shall be performed on units which have been subjected to, and have passed, the production inspection specified in 4.4.2.

TABLE V. Periodic inspection.

Inspection	Requirement paragraph	Test paragraph
Interference Electromagnetic field	3.4.9 3.4.10	4.5.6 4.5.6

4.4.3.1 Sampling for periodic inspection. One transmitter shall be selected from each successive 50 transmitters, or fraction thereof, produced. The first sample transmitter shall be selected from the first month's production.

4.4.3.2 Nonconforming periodic sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

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4.4.3.3 Reinspection of conforming periodic sample units. Unless otherwise specified in the contract (see 6.2), sample units which have been subjected to, and have passed, periodic tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after any necessary rework.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.6.

4.5.1 MIL-E-16400 tests. When 4.5.1 is specified in TABLE IV, the test specified in MIL-E-16400 shall be applied.

4.5.2 Test by analysis. When 4.5.2 is specified in TABLE IV, the test may be performed by analysis or calculation, or both.

4.5.3 Cables. Cables to be supplied with the end item that are not captive to the transmitter shall, as an assembly, be subjected to a continuity and voltage test. The continuity test shall ensure that wires terminate on the correct connector pins. The voltage test shall identify any leakage between conductors. The voltage to be applied to each wire or cable for the voltage test shall be equal to the highest voltage rating, without exceeding the voltage rating for any constituent part of the cable assembly.

4.5.4 Leakage current test. The leakage current test shall be as specified in 4.5.4.1 and 4.5.4.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE TRANSMITTER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

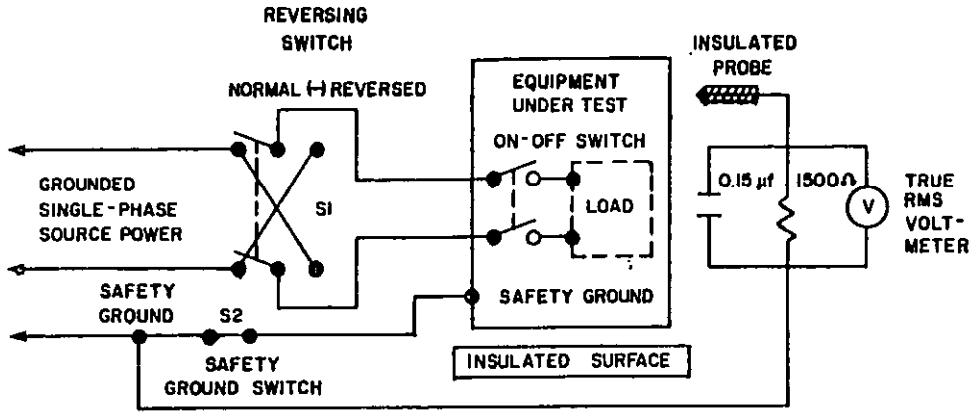
4.5.4.1 Transmitter connections. Each transmitter directly connected to an external power source and units deriving power from the transmitter shall be placed on an insulated surface. All safety ground conductors between the transmitter and units deriving power from the transmitter shall be intact. The safety ground conductor between the transmitter under test and the source power shall be opened during the test. The transmitter shall be connected as shown in FIGURE 3, if it is connected to single-phase source power, and as shown in FIGURE 4, if it is connected to three-phase source power.

4.5.4.2 Measurement. Leakage current shall be measured on the transmitter in its normal operating configuration. Transmitter controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the transmitter is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURES 3 and 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.5 Contractor validation. The contractor shall inspect the transmitter to determine technical compliance and adequacy for the specified characteristic or requirement. This validation shall include qualitative examination and testing whenever a function is specified.

4.5.6 Inspection by testing. When this paragraph is specified, the contractor shall perform quantitative testing.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OF. REPEAT STEP 4.
6. S1 SW REVERSED
7. ON-OFF SW ON. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

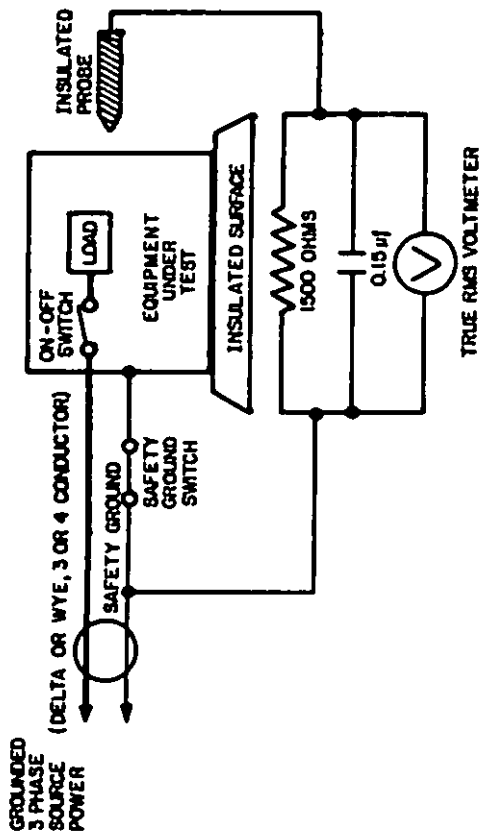
WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 3. Single-phase test diagram for leakage current measurement.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. SAFETY GROUND SWITCH CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING. SAFETY GROUND SWITCH OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. SAFETY GROUND SWITCH CLOSED.
7. REPEAT STEP 3 THRU 6 FOR EACH MODE OF OPERATION.
8. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

NOTE: 1. ALL THREE PHASES SHALL BE CONNECTED DURING MEASUREMENT.
2. THE SAFETY GROUND CONDUCTOR SHALL NOT CARRY LOAD CURRENT.

WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 4. Three-phase test diagram for leakage current measurement.

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4.6 Reliability verification test. A reliability verification test shall be performed in accordance with Test Plan XXC of MIL-STD-781 on the two first article units. Environmental and operational duty cycles shall be as specified herein.

4.6.1 Environmental conditions for reliability testing. The environmental conditions shall be as specified in a through d:

a. Input voltage: Nominal ±10 percent
b. Vibration: None
c. Temperature: +50°C, +5°C, -0°C
d. Moisture: 75 percent RH

4.6.2 Performance tests. At least daily, performance measurements shall be made to verify the performance characteristics specified in a through d:

a. Power out
b. Frequency and long term frequency stability
c. Modulation
d. Intermodulation

4.6.3 Test frequencies. The test frequencies specified in a through d shall be used:

a. F_1: 2.0005 MHz
b. F_2: 4.9995 MHz
c. F_3: 10.0019 MHz
d. F_4: 29.9999 MHz

4.6.4 Test schedule. The test shall consist of three 8-hour schedules per day, the first being manned, where performance measurements shall be made. The other two schedules may be unmanned and pertinent data may be automatically recorded to the extent that instrumentation permits. These schedules shall be as specified in a and b:

a. Manned schedule:
   1. 0 hours to 2 hours: Transmitter on
   2. 2 hours to 3 hours: Transmitter performance measurements
   3. 3 hours to 4 hours: Transmitter standby
   4. 4 hours to 6 hours: Transmitter on
   5. 6 hours to 7 hours: Transmitter measurements
   6. 7 hours to 8 hours: Transmitter standby
b. Unmanned schedule:
   1. 0 hours to 3 hours: Transmitter on
   2. 3 hours to 4 hours: Transmitter standby
   3. 4 hours to 7 hours: Transmitter on
   4. 7 hours to 8 hours: Transmitter standby

4.6.5 Transmitter schedules. The mode of operation and frequency of transmission for each transmitting ON period of the manned schedules shall occur in the sequence specified herein.

Day	Frequency	Mode of transmission (into a 50-ohm dummy load)
1	F_1: First 4 hours of manned schedule F_2: Remainder of the day	Amplitude modulation (A3E), 50 percent rated avg carrier 30 percent to 50 percent, 1000 Hz modulation; transmitting for 10 minutes out of each 15-minute period.
2	F_3: First 4 hours of manned schedule F_4: Remainder of the day	CW (A1), full rated avg power keyed at a speed of 25 dot Hz transmitting for 10 minutes out of each 15-minute period.
3	F_1: First 4 hours of manned schedule F_2: Remainder of the day	FSK (F1), full rated avg power keyed from space to mark at a 30 Hz rate (RYs); transmitting for 10 minutes out of each 15-minute period
4	F_3: First 4 hours of manned schedule F_4: Remainder of the day	ISB (A3B), 100 percent of rated PEP, two-tone modulation per sideband transmitting for 10 minutes out of each 15-minute period.

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The same sequence shall be reused on the next four days, except that the sequence of frequencies shall be reversed; F_4 and F_3 on the fifth day, F_2 and F_1 on the sixth day, F_4 and F_3 on the seventh day, and F_2 and F_1 on the eighth day. These schedules shall be repeated as necessary.

4.7 Maintainability demonstration. The contractor shall demonstrate that the transmitter conforms to the maintainability requirements specified in 3.9.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.2.2.

4.8.1 General workmanship. The transmitter, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.10.1. Each solder connection and its associated wiring or leads shall be visually examined.

4.8.2 Workmanship screen. Vibration (see 4.8.2.1) and temperature cycling (see 4.8.2.2) shall be performed on each equipment. Vibration shall be performed prior to temperature cycling. The vibration shall be performed at the assembly or drawer level. The cabinet, with drawers removed, shall not require vibration.

4.8.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to the vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 5. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the avg of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the acceleration level. When the RF amplifier and power supply are vibrated, the driver tubes in the RF amplifier may be removed and the high voltage from the power supply may be turned off. All other operating voltages shall be applied. Appropriate input signals shall be applied as necessary to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

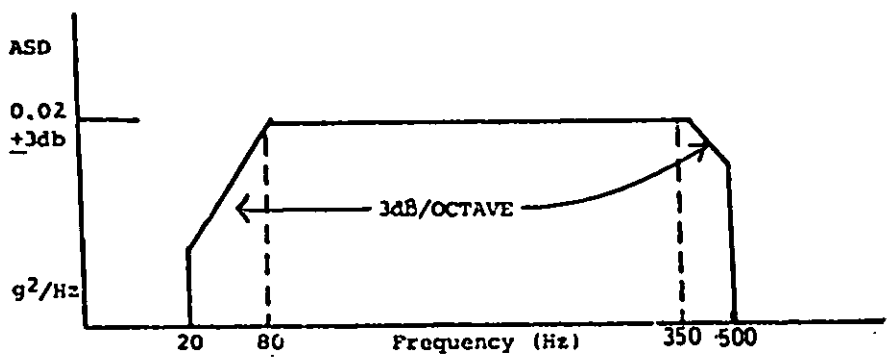


FIGURE 5. Random vibration curve.

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4.8.2.2 Temperature cycling. Each unit shall be subjected to 10 cycles of the temperature curve shown in FIGURE 6. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. When performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part of the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

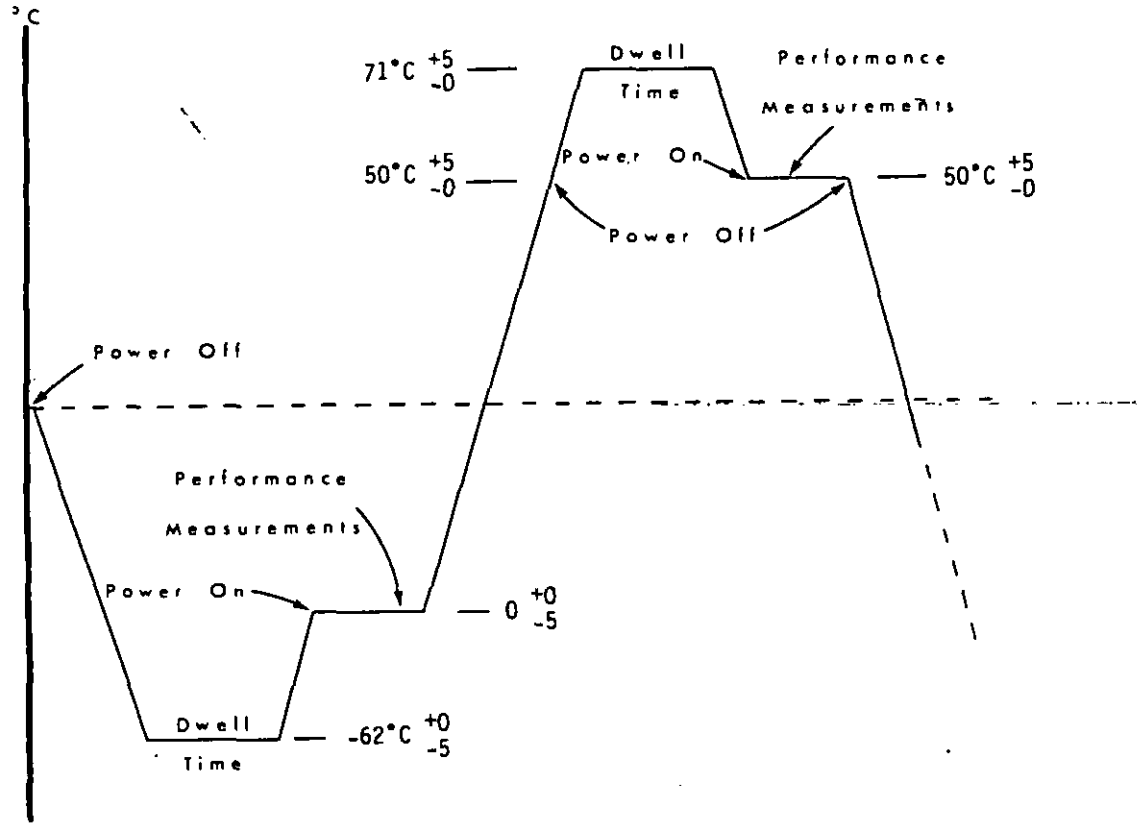


FIGURE 6. One cycle of temperature curve (not to scale).

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4.9 Inspection of preparation for delivery. Inspection shall be conducted to ensure compliance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The transmitter covered by this specification is a Navy general-purpose shore based radio transmitter for point-to-point HF communications.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Use of test equipment not specified in MIL-STD-1364 (see 3.7.2)
c. Number of FAS to be submitted if other than specified in 4.3
d. When reinspected periodic sample units may not be accepted on the contract (see 4.4.3.3)
e. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulation. The first article should be a first production item. The first article should consist of two units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.5.

6.4.1 Long term overvoltage and undervoltage. Long term overvoltage and undervoltage are defined as voltage excursions which are above or below the nominal operating voltage and within 10 percent of the nominal voltage.

6.4.2 Electrical noise. Electrical noise is defined as any signal present in the transmitter output whose frequency is within ±6.3 kHz of the carrier frequency when the transmitter is conditioned for normal operation with the four audio inputs terminated resistively in their characteristic impedance of 600 ±j0 ohms and no input signal is applied.

6.4.3 Absolute frequency error. Absolute frequency error is defined as the total deviation in Hz of the output carrier frequency (present in the output or fully suppressed) from that frequency indicated on the exciter frequency selection controls or their associated indicating devices.

6.4.4 Frequency calibration. Frequency calibration of the exciter is defined as a process for:

a. Determining the frequency stability
b. Determining the absolute frequency error
c. Reducing the absolute frequency error to less than 0.3 Hz at any frequency covered by the transmitter

6.4.5 Exciter standby condition. Exciter standby condition is defined as that condition wherein the maximum amount of power consuming circuitry in the exciter is deactivated while all circuitry which affects the stability and accuracy of the exciter output frequency remains in full operating condition.

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6.5 Release for shipment. Release for shipment shall be as specified in 6.5.1.

6.5.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the periodic inspection.

6.6 GFE. The contracting officer should arrange to furnish the equipment listed in 3.2.12.

6.7 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.8 Supersession data. This specification includes the requirements of ELEX-T-404, dated 14 March 1980.

Preparing activity:
NAVY - EC

(Project 5820-N780)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-28885	Transmitter, Radio, AN/FRT-83(B), 1000 Watts Peak Envelope Power and Average, HF

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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